

Untitled Space Craft

Complete Documentation

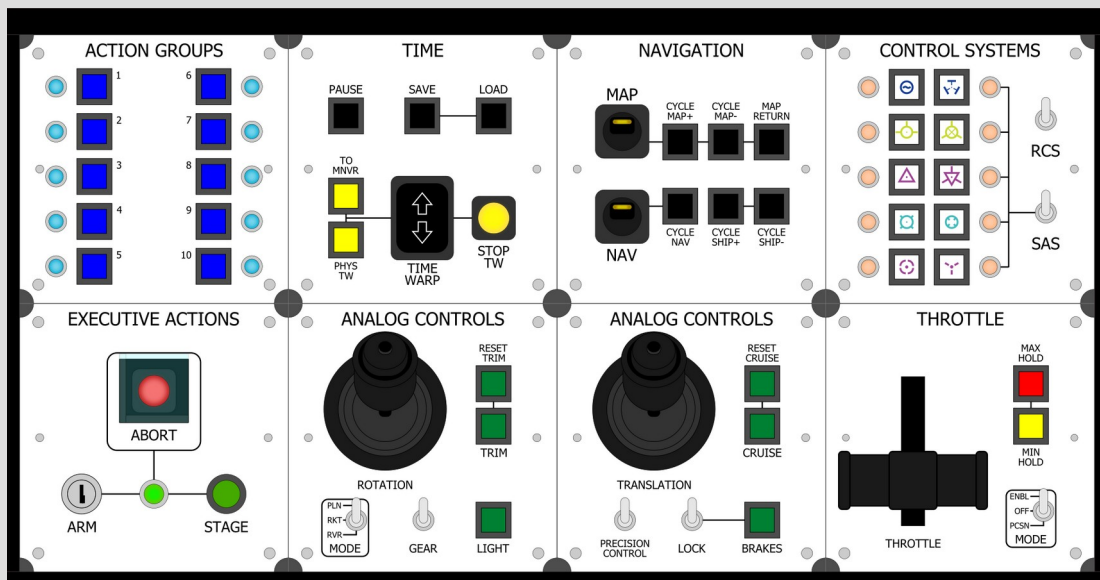


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Introduction

Untitled Space Craft is a brand of game controllers specialized for playing Kerbal Space Program and its derivatives, created by Codapop in 2021. This project is open source.

This document serves as a guide for those who choose to build their own USC controller.

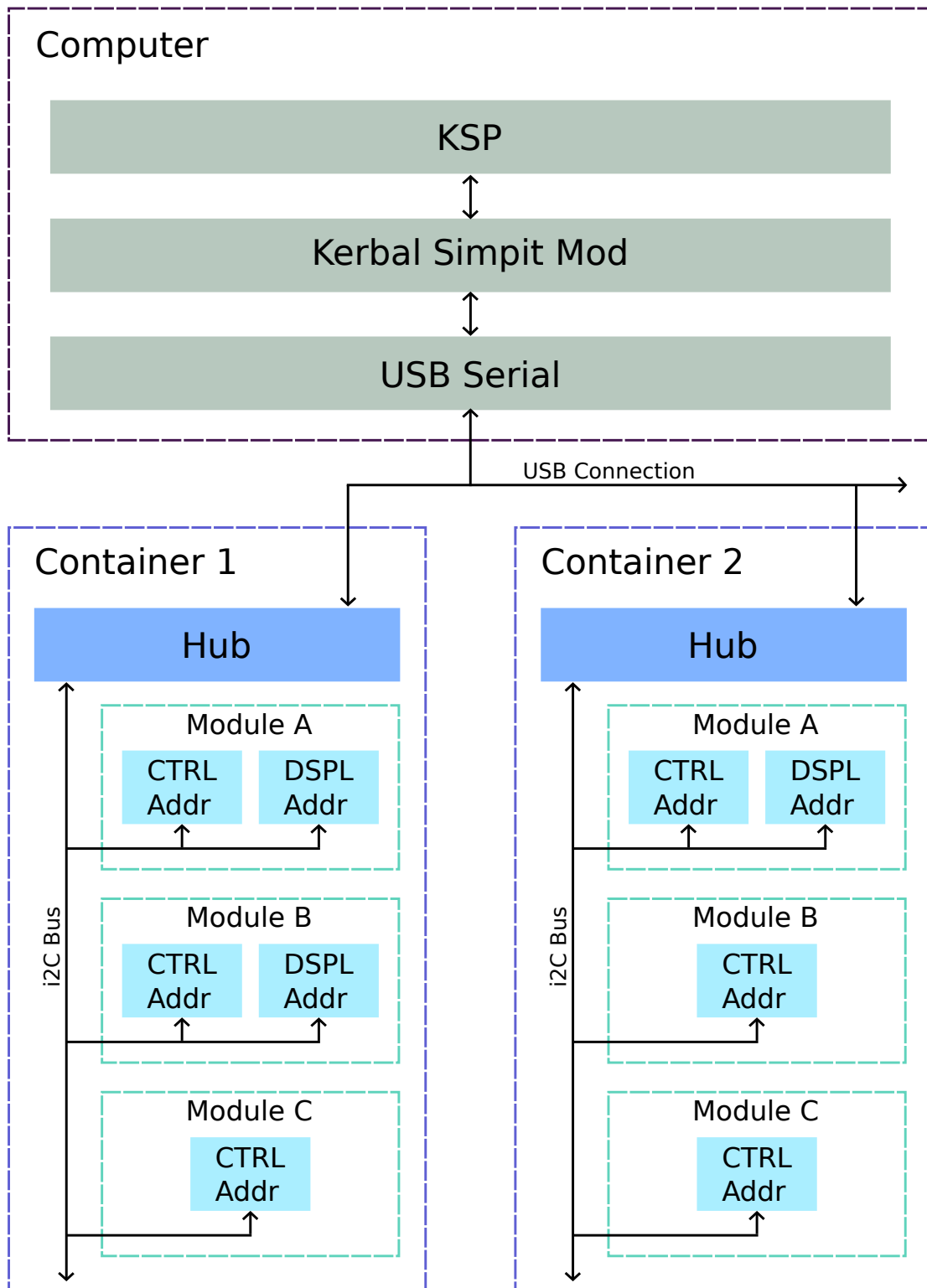
The USC ecosystem is designed around a modular architecture. Individual control panels, referred to as *modules*, connect to a central *Hub* and communicate over a shared I²C bus. This allows builders to create anything from a small desktop controller with a single module to a full cockpit consisting of multiple containers and dozens of independent controls.

This manual covers the hardware, electronics, firmware, and assembly process required to build a USC controller. It includes mechanical drawings, schematics, bills of materials, firmware architecture, and assembly instructions. While basic soldering and electronics experience is recommended, every effort has been made to document the project thoroughly so that builders can understand not only how to assemble the controller, but also how it works and how to modify or extend it.

System Diagram

Untitled Space Craft utilizes the [Kerbal Simpiti Revamped](#) mod for KSP in order to communicate with the game. The mod talks to controllers over the serial bus. Multiple containers can be connected to Simpiti by editing the `settings.cfg` file in the `GameData/KerbalSimpiti` folder.

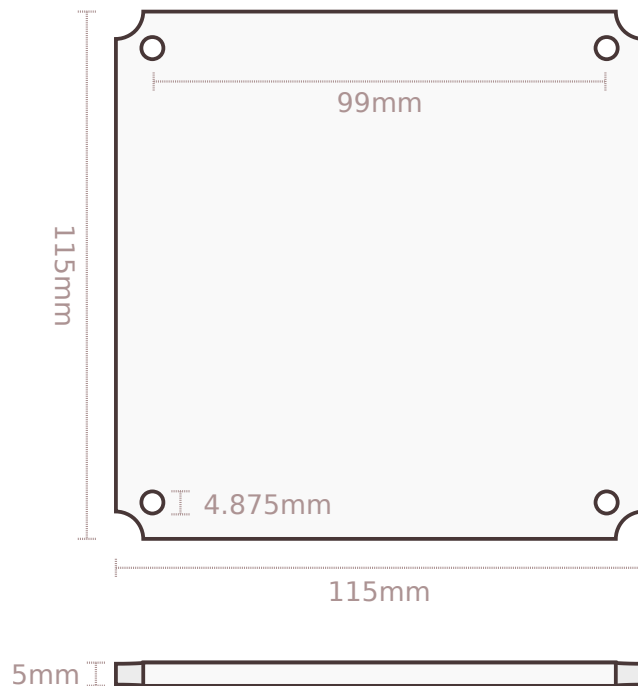
Each container has a Hub, powered by Arduino Micro, with up to 8 modules driven by ATtiny microcontrollers. They communicate via the i2C protocol. Each module has one (for *just control*) or two (for *control and display*) addresses, designated with **CTRL** and **DSPL**.



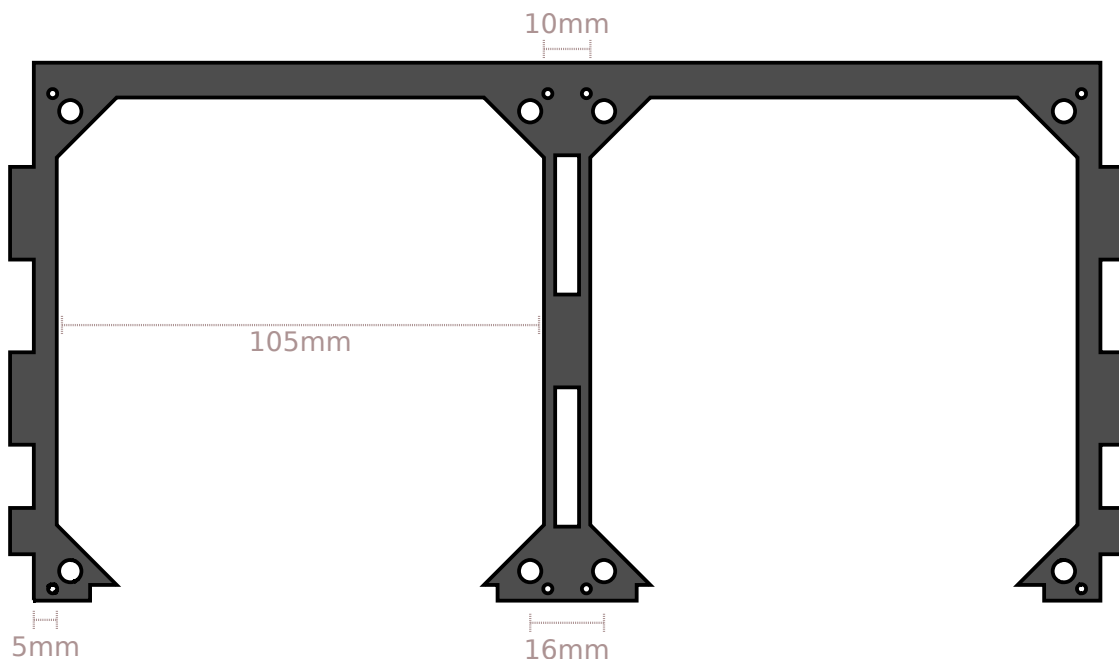
The Standard Module

The standard module is the form factor that governs the design of USC. Each module follows this same basic body plan.

It is 115mm square, with holes that are 4.875mm in diameter and 99mm apart from their centers, and 5mm thick. The hole diameter is a function of the diameter of the neodymium magnets, the flexibility of the medium (typically acrylic), and the thickness and settings of the laser. It is advised to do a test cut first to guarantee a proper friction fit.

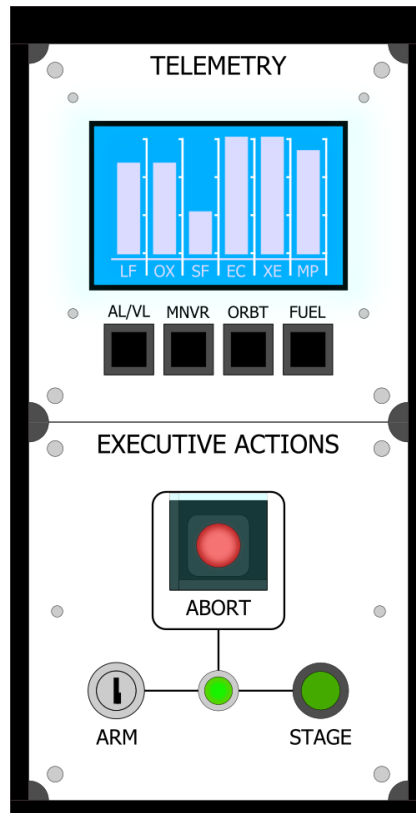


The module fits into the Module Slot. They have the same holes for magnets, 16mm apart where the modules touch. The slots are 105mm square, with a 5mm edge (or 10mm between them).

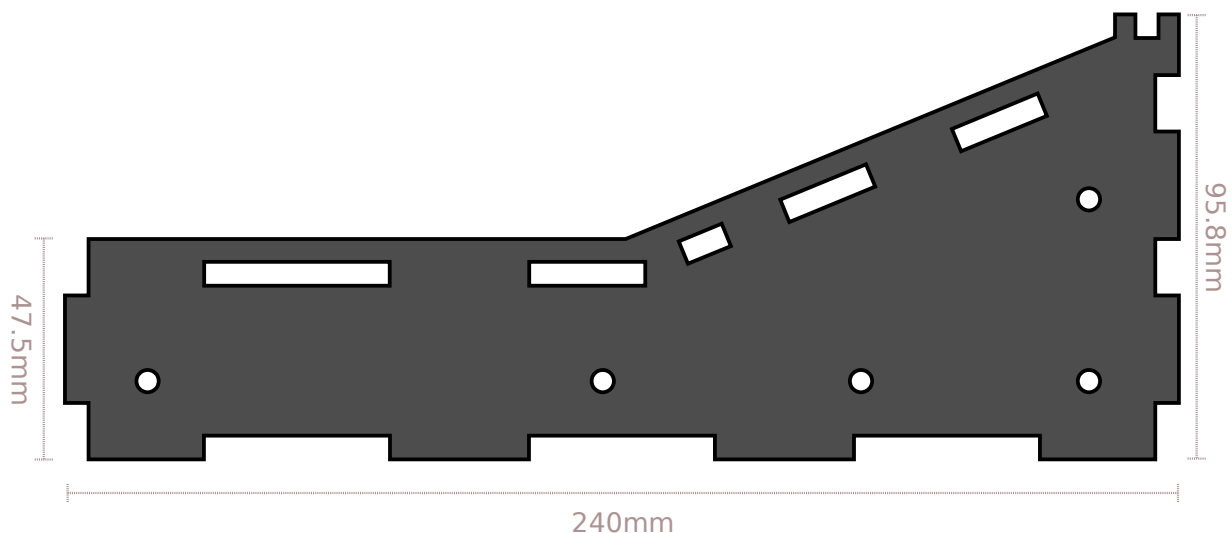


Containers

A container houses one or more standard modules and provides power and communication through the internal data bus.



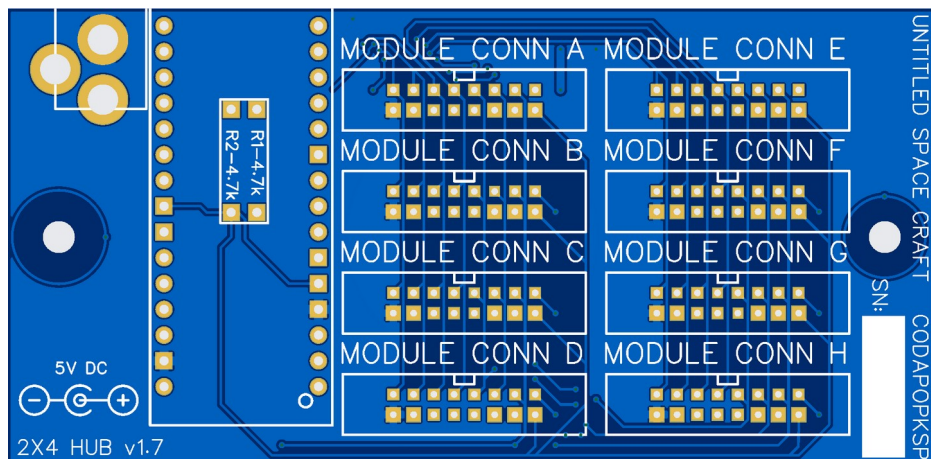
Containers have several measurements in common. **Their pieces should all be 5mm thick.** They are all 95.8mm tall on the back end. Containers that are two modules deep are 240mm long and 47.5mm tall on the short end. The measurements throughout the containers may not be exact; typically they are a few fractions of a millimeter shorter in order to account for the space needed for the caulk adhesive.



Hubs

Every container has exactly one Hub. A Hub is the core of the controller and supplies the serial connection to the computer and contains KSP-specific firmware.

A single hub consists of a consumer-grade microcontroller, typically Arduino Micro (or Arduino Nano on older USC models) as well as either four or eight connections to the i2c Data Bus to connect modules.



There are two sizes of hubs, Quad (aka: 1x4) and Octo (aka: 2x4). The two hub sizes are identical in every way other than the number of sockets that connect to the i2c bus. The 2.1mm barrel jack is center-positive polarity. The hub PCB is set up to support both Arduino Micro and Arduino Nano interchangeably, but the firmware expects Arduino Micro in order to make use of the HID capability for keyboard emulation (both in KSP Mode and in Emulation Mode).

The microcontroller is powered by the USB connection while the modules are powered by an external 5V PSU that connects via the 2.1mm barrel jack.



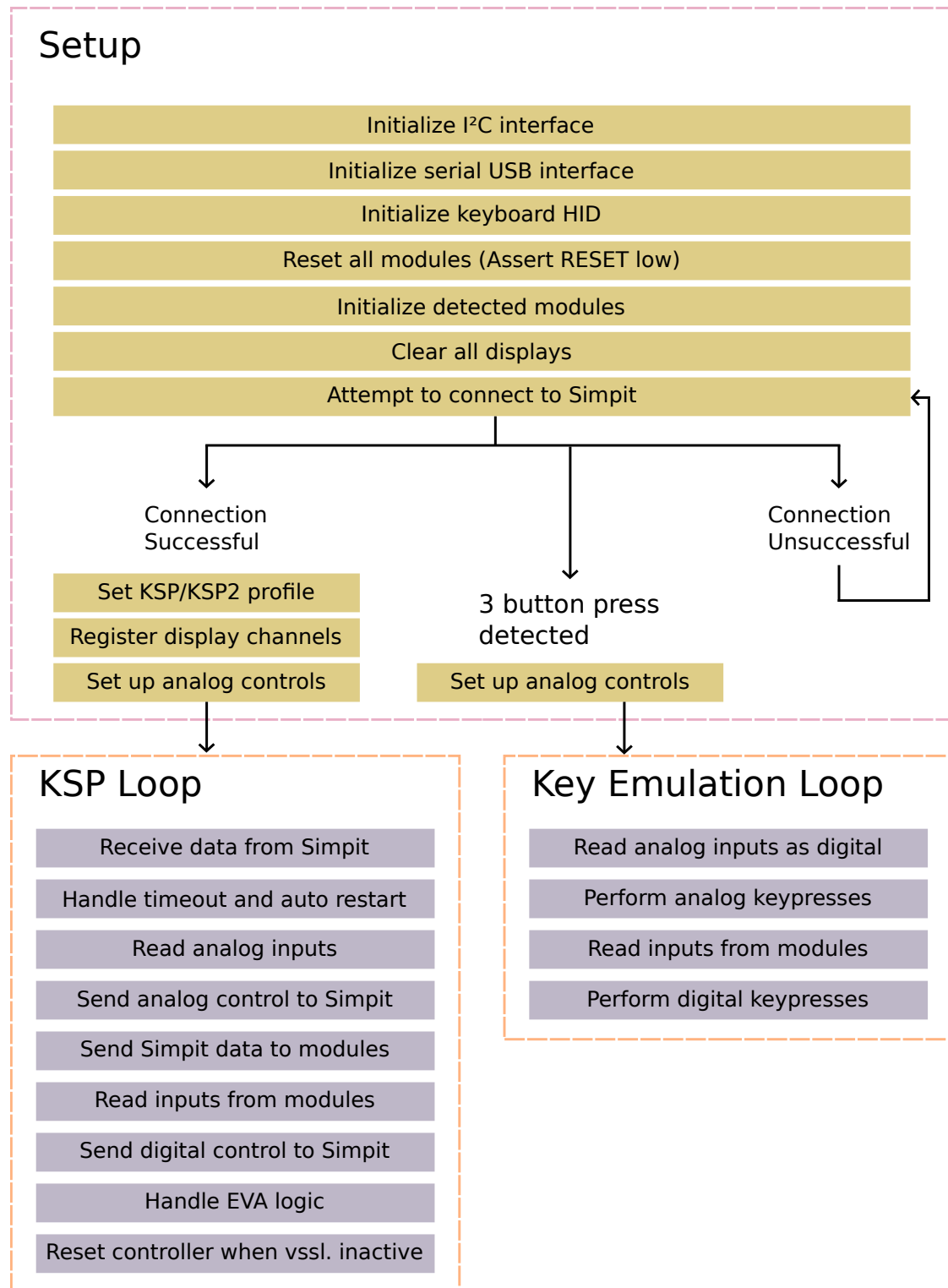
The microcontroller should not be soldered directly to the PCB; it should be socketed with two 1x15 pin header sockets, increasing the height by approximately 8.5mm in order to match the holes in the back of the container.

The PCB itself should have a gap of one standard 3M nut between it and the acrylic below it.

Firmware Execution Flow

Untitled Space Craft firmware is written in C/C++ and targets the Arduino platform. On power-up, the Hub initializes the hardware, discovers connected modules, and attempts to establish communication with Kerbal Simpit.

Once it is ready, it will repeatedly attempt to connect to Simpit for either KSP1 or KSP2. During this time, if 3 repeated inputs are detected, it will enter Keyboard Emulation mode. If the connection to Simpit is successful, it will enter KSP mode.



Module Firmware

Each module has one or two microcontrollers with firmware that manages their module's set of inputs and outputs. Typically, these are specialized based on Control (CTRL) or Display (DSPL), meaning buttons/switches or LED outputs respectively. The only alternative is the Telemetry module which runs an Arduino Nano that handles both as well as the screen drawing functionality.

Both types of firmware utilize the i2c protocol and use the TinyWire library.

CTRL

The CTRL firmware uses the ShiftIn library to collect inputs, as bits, which it stores in bytes. When requested by the master (Wire.requestFrom(ctrl_address, in_bytes);), it provides one or two bytes.

```
void onI2CRequest() {
    TinyWire.send(bytes[0]);
    TinyWire.send(bytes[1]); //Optional
}
```

DSPL

The DSPL firmware uses the ShiftRegister74HC595 library to drive LED arrays. When it receives data from the master in the form of bytes, it converts them into serial bits.

```
void onI2CReceive(int howMany) {
    bytes[0] = TinyWire.read();
    bytes[1] = TinyWire.read(); //Optional
}
```

Analog

In the Analog firmware, digital inputs are collected and condensed into bytes in the same manner as CTRL firmware. Each joystick axis is read using the microcontroller's ADC, producing a 10-bit value. Because this value cannot fit into a single byte, each axis is transmitted as two bytes containing the high and low portions of the ADC value. It also sends a byte of other input data.

```
axis1 = analogRead(A1);
axis2 = analogRead(A2);
axis3 = analogRead(A3);

void onI2CRequest() {
    TinyWire.send(bytes[0]); // Other inputs
    TinyWire.send(highByte(axis1));
    TinyWire.send(lowByte(axis1));
    TinyWire.send(highByte(axis2));
    TinyWire.send(lowByte(axis2));
    TinyWire.send(highByte(axis3));
    TinyWire.send(lowByte(axis3));
}
```

I²C Addresses

Module	Address
Special	0-7
	8
	9
Stage	10
Abort	11
Executive Actions	12
Camera	13
Navigation	14
Navigation (Time)	15
Executive Actions (Groups) – Ctrl	16
Executive Actions (Groups) – Dspl	17
Executive Actions (Control) – Ctrl	18
Executive Actions (Control) – Dspl	19
Time – Ctrl	20
Time – Dspl	21
EVA - Ctrl	22
EVA – Dspl	23
Control Systems – Ctrl	24
Control Systems – Dspl	25
Utility Navication	26
Utility Time	27
	28
	29
Action – Ctrl	30
Action – Dspl	31
Action 2 – Ctrl	32
Action 2 – Dspl	33
Action 3 – Ctrl	34
Action 3 – Dspl	35
	36
	37
	38
	39
Telemetry	40
	41
	42
	43
	44
Analog (Throttle)	45
Rotation (Throttle)	46
Translation	47
Analog	48
Rotation	49
Throttle	50
	51
	52
	53
	54
Fuel – Ctrl	55
Fuel – Dspl A	56
Fuel – Dspl B	57
	58
	59
Velocity – Ctrl	60
Velocity – Dspl	61
Radar – Ctrl	62
Radar – Dspl	63
Apsides – Ctrl	64
Apsides – Dspl	65

Module	Address
	66
	67
	68
	69
System – Ctrl	70
System – Dspl/Snd	71
	72
	73
	74
	75
	76
	77
	78
	79
	80
	81
	82
	83
	84
	85
	86
	87
	88
	89
Incompatible	90-119
Special	120-127

Purpose	Range
Input/Output	10 – 44
Analog	45 – 54
Display	55 – 89

Version
1.0
2.0
3.0
4.0
5.0
6.0

Bill of Materials

All parts are through-hole, 2.54mm pitch unless otherwise stated.

Links are provided for convenience and may not correspond exactly to the recommended part. Always verify dimensions and specifications against the datasheet before ordering.

Module	Part	Qty	Value	Note
All Modules	2.0mm Pitch 16pin Connector	1	Socket	Standard module connector
	Electrolytic Cap	1	10uf 16v	Main power decoupling
	Round Neodymium Magnets	4	5mm/5mm	Requires high precision
	PCB	1		
	Acrylic Face Plate	1		
	Screw	2	3M, 15mm	Telemetry needs 4
	Nut	4	3M	Telemetry needs 16
Abort	Circle Button	1	Red, 12mm	OFF-(ON)
	Safety Switch	1	Red	
	LED	1	Red, 10mm	
	Chip Socket	1	8 Pin	
	Resistor	1	1.2K, 1/4W	
	Resistor	1	10k, 1/4W	
	Ceramic Capacitor	1	100nf	Type 104
Action Groups	Attiny25	1		
	Square Button	10	Blue	
	LED	10	Blue, 3mm	
	LED Socket	10	3mm	
	Chip Socket	2	8 Pin	
	Chip Socket	4	16 Pin	
	Resistor	10	10k, 1/4W	
	Resistor	10	62k, 1/4W	
	Ceramic Capacitor	6	100nf	Type 104
	Attiny45	2		
	74HC165	2		
Analog	74HC595	2		
	Square Button	2	Blue	
	Switch	2	SPST	
	Switch	1	SPDT 3-Pos	ON-OFF-ON
	JH-D400X Joystick	1		
	Chip Socket	1	14 Pin	
	Chip Socket	1	16 Pin	
	Resistor	7	10k, 1/4W	
	Ceramic Capacitor	2	100nf	Type 104
	Attiny24	1		
Analog Throttle	74HC165	1		
	Square Button	2	Blue	
	Switch	1	SPST	
	Switch	1	SPDT 3-Pos	ON-OFF-ON
	JH-D400X Joystick	1		
	Chip Socket	1	14 Pin	
	Chip Socket	1	16 Pin	
	Resistor	6	10k, 1/4W	
	Ceramic Capacitor	2	100nf	Type 104
	Attiny24	1		
	74HC165	1		
	Potentiometer	1	B20K	
Camera	Potentiometer Cover Dial	1		
	Square Button	4	Black	
	Square Button	6	White	
	Chip Socket	1	8 Pin	
	Chip Socket	2	16 Pin	
	Resistor	10	10k, 1/4W	
	Ceramic Capacitor	3	100nf	Type 104
	Attiny45	1		
	74HC165	2		

Module	Part	Qty	Value	Note
Control Systems	Square Button	10	White	
	Switch	2	SPST	
	LED	10	Orange, 3mm	
	LED Socket	10	3mm	
	Chip Socket	2	8 Pin	
	Chip Socket	4	16 Pin	
	Resistor	22	10k, 1/4W	
	Ceramic Capacitor	6	100nf	Type 104
	Attiny45	2		
	74HC165	2		
	74HC165	2		
Executive Actions (Control)	Square Button	1	White	
	Square Button	2	Blue	
	Circle Button	1	Green, 12mm	OFF-(ON)
	LED Button	1	Red, 16mm	
	Button Safety Cover	1	22x24mm	
	Switch	2	SPST	
	Key Switch	1	12mm	
	LED	3	Orange, 3mm	
	LED	2	Blue, 3mm	
	LED Socket	5	3mm	
	Chip Socket	2	8 Pin	
	Chip Socket	2	16 Pin	
	Resistor	1	1.2K, 1/4W	
	Resistor	10	10k, 1/4W	
	Resistor	2	62k, 1/4W	
	Ceramic Capacitor	4	100nf	Type 104
	Attiny45	2		
	74HC595	1		
	74HC165	1		
Executive Actions (Groups)	Square Button	5	Blue	
	Circle Button	1	Green, 12mm	OFF-(ON)
	LED Button	1	Red, 16mm	
	Button Safety Cover	1	22x24mm	
	Key Switch	1	12mm	
	LED	5	Blue, 3mm	
	LED Socket	5	3mm	
	Chip Socket	2	8 Pin	
	Chip Socket	2	16 Pin	
	Resistor	1	1.2K, 1/4W	
	Resistor	7	10k, 1/4W	
	Resistor	5	62k, 1/4W	
	Ceramic Capacitor	4	100nf	Type 104
	Attiny45	2		
	74HC595	1		
	74HC165	1		
Executive Actions	Circle Button	1	Green, 12mm	OFF-(ON)
	LED Button	1	Red, 16mm	
	Button Safety Cover	1	24x26mm	
	Key Switch	1	12mm	
	LED	1	Green, 5mm	
	LED Socket	1	5mm	
	Chip Socket	1	8 Pin	
	Resistor	1	300, 1/4W	
	Resistor	1	1.2K, 1/4W	
	Resistor	2	10k, 1/4W	
	Ceramic Capacitor	1	100nf	Type 104
	Attiny25	1		

Module	Part	Qty	Value	Note
EVA	Square Button	6	White	
	Square Button	5	Green	
	LED Array	1	Green, 10 LED	
	Chip Socket	2	8 Pin	
	Chip Socket	4	16 Pin	
	Resistor	10	1.2K, 1/4W	
	Resistor	11	10k, 1/4W	
	Ceramic Capacitor	6	100nf	Type 104
	Attiny45	2		
	74HC595	2		
	74HC165	2		
Navigation	Square Button	6	Black	
	LED Switch	2	Yellow	
	Chip Socket	1	8 Pin	
	Chip Socket	1	16 Pin	
	Resistor	2	1.5k, 1/4W	
	Resistor	8	10k, 1/4W	
	Ceramic Capacitor	2	100nf	Type 104
	Attiny45	1		
	74HC165	1		
Natigation (Time)	Square Button	2	Black	
	Square Button	2	Yellow	
	LED Button	1	Yellow, 16mm	
	D36 LED Switch	1	Yellow	
	6P Rocker Switch	1		
	Chip Socket	2	8 Pin	
	Chip Socket	1	16 Pin	
	Resistor	1	300, 1/4W	
	Resistor	1	1.5k, 1/4W	
	Resistor	8	10k, 1/4W	
	Ceramic Capacitor	3	100nf	Type 104
	Attiny45	2		
	74HC165	1		
Rotation	Square Button	3	Green,	
	Switch	1	SPST	
	Switch	1	SPDT 3-Pos	ON-OFF-ON
	JH-D400X Joystick	1		
	Chip Socket	1	14 Pin	
	Chip Socket	1	16 Pin	
	Resistor	6	10k, 1/4W	
	Ceramic Capacitor	2	100nf	Type 104
	Attiny24	1		
	74HC165	1		
Rotation (Throttle)	Square Button	2	Green,	
	Switch	1	SPST	
	Switch	1	SPDT 3-Pos	ON-OFF-ON
	JH-D400X Joystick	1		
	Chip Socket	1	14 Pin	
	Chip Socket	1	16 Pin	
	Resistor	4	10k, 1/4W	
	Ceramic Capacitor	2	100nf	Type 104
	Attiny24	1		
	Potentiometer	1	B20K	
	Potentiometer Cover Dial	1		

Module	Part	Qty	Value	Note
Stage	Circle Button	1	Green, 12mm	OFF-(ON)
	Safety Switch	1	Green,	
	LED	1	Green, 10mm	
	Chip Socket	1	8 Pin	
	Resistor	1	150, 1/4W	
	Resistor	1	10k, 1/4W	
	Ceramic Capacitor	1	100nf	Type 104
	Attiny25	1		
Telemetry	Square Button	4	Black	
	LCD Screen	1	128x64	ST7920 driven
	Header Pins	1	1x20 Pin	Not socket
	Resistor	4	10k, 1/4W	
	Washer	8	3M	Depends on thickness
	Arduino Nano	1		
Throttle	Square Button	1	Yellow	
	Square Button	1	Red	
	Switch	1	SPDT 3-Pos	ON-OFF-ON
	Potentiometer	1	B20k, 6 Pin	Two parallel channels
	Throttle Lever	1		3D printed
	Throttle Friction Mechanism	1		3D printed
	Chip Socket	1	8 Pin	
	Resistor	2	10k, 1/4W	
	Diode	1	1N4148	
	Ceramic Capacitor	1	100nf	Type 104
	Attiny25	1		
Time	Square Button	3	Black	
	Square Button	2	Yellow	
	LED Button	1	Yellow, 16mm	
	6P Rocker Switch	1		
	Chip Socket	2	8 Pin	
	Chip Socket	1	16 Pin	
	Resistor	1	300, 1/4W	
	Resistor	8	10k, 1/4W	
	Ceramic Capacitor	3	100nf	Type 104
	Attiny45	2		
	74HC165	1		
Translation	Square Button	3	Green,	
	Switch	2	SPST	
	JH-D400X Joystick	1		
	Chip Socket	1	14 Pin	
	Resistor	5	10k, 1/4W	
	Ceramic Capacitor	1	100nf	Type 104
	Attiny24	1		
Utility (Navigation)	Square Button	2	Black	
	Square Button	5	Green	
	D36 LED Switch	1	Yellow	
	LED Array	1	Green, 10 LED	
	Chip Socket	2	8 Pin	
	Chip Socket	3	16 Pin	
	Resistor	10	1.2K, 1/4W	
	Resistor	1	1.5k, 1/4W	
	Resistor	8	10k, 1/4W	
	Ceramic Capacitor	5	100nf	Type 104
	Attiny45	2		
	74HC595	2		
	74HC165	1		

Module	Part	Qty	Value	Note
Utility (Time)	Square Button	3	Black	
	Square Button	5	Green	
	LED Array	1	Green, 10 LED	
	Chip Socket	2	8 Pin	
	Chip Socket	3	16 Pin	
	Resistor	10	1.2K, 1/4W	
	Resistor	8	10k, 1/4W	
	Ceramic Capacitor	5	100nf	Type 104
	Attiny45	2		
	74HC595	2		
	74HC165	1		
Quad Hub	2.0mm Pitch 16pin Connector	4	Socket	Standard module connector
	Resistor	2	4.7k, 1/4W	
	Nut	6	3M	
	Arduino Micro	1		Can be substituted with a Nano
	Header Sockets	2	15 Pin, Socket	For the arduino to slot into
	Barrel Jack	1	2.1mm	
Octo Hub	2.0mm Pitch 16pin Connector	8	Socket	Standard module connector
	Resistor	2	4.7k, 1/4W	
	Nut	6	3M	
	Arduino Micro	1		Can be substituted with a Nano
	Header Sockets	2	15 Pin, Socket	For the arduino to slot into
	Barrel Jack	1	2.1mm	
Miscellaneous	Power Supply	1	5V	Center 5V, Sleeve ground
	Micro USB cable	1		
	Neodymium Magnets	10	5mm/5mm	
	Ribbon Cable	2-3m	16 wire, 1mm	Depending on size
	Ink	1	Black	Alcohol or acetone based
	Acrylic Container	1		Depending on size
	2.0mm Pitch 16pin Connector	2-16	Pin	Two per module

Required Tools

- Soldering iron and all relevant materials (solder, flux, etc.)
- Wire (standard hobby gauge, ~22-24 AWG)
- Wire strippers/cutters
- Pliers
- Screwdriver, with different size tips
- Vise/vice or some reliable way to compress objects
- 8mm wrench (for all switches and 3mm LEDs)
- 9mm wrench (Analog Throttle and Rotation Throttle modules)
- 15mm wrench (Executive Actions, Stage, Abort modules)
- Ink (for lettering, black or your choice)
- A suitable way to clean the ink (such as acetone-based nail polish remover)
- Silicone caulk (clear or the color of your container)
- A breadboard
- Arduino IDE software or similar
- Multimeter

Module Assembly

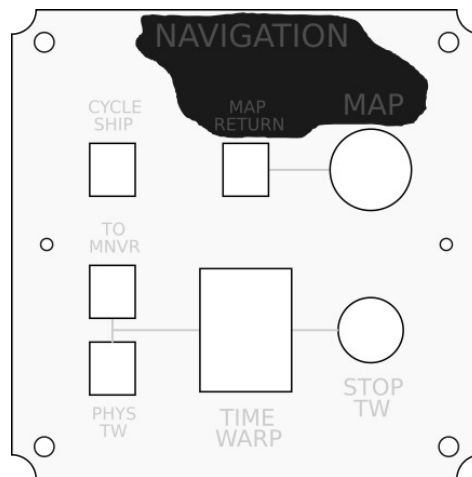
Untitled Space Craft modules follow a consistent manufacturing design language, the process of which can be separated into three parts: Faceplate fabrication, component placement, and PCB soldering.

Faceplate Fabrication

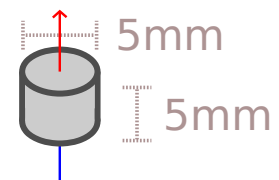
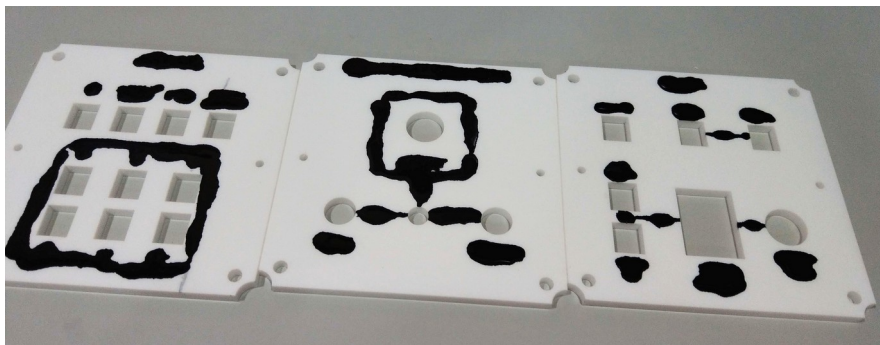
These instructions are for if you are using acrylic for the faceplates. If you are using another material, then you will have to modify these instructions to fit your use case.

Faceplates should be laser cut from 5mm acrylic. A suitably powerful laser should be used to prevent the resulting cuts from becoming a pyramid shape. The faster the cut the less the surrounding material gets burned by an unfocused laser.

Inking the faceplates involves liberally applying ink to the etched lines so that the ink fills all the gaps. It is recommended to clean the burnt acrylic dust out of the gaps first.



Any solvent-based ink can be used, such as alcohol-based or acetone-based, typically used in dry-erase markers or permanent markers. Inking can be performed with or without the paper covering. After applying the ink, wipe the excess off with a tissue and wait for the rest to dry. Once it has dried, you can remove any remaining excess using the same solvent as your ink.



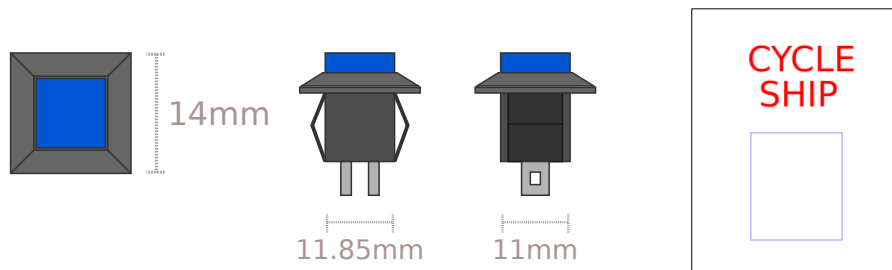
Magnets can be added to the acrylic by pressing them in with a vise or similar compression tool. Magnets should be inserted with the **NORTH** side **UP** and pushed in from the top side to ensure the side that contacts the container is flush. Each magnet is 5mm in diameter and 5mm thick.

Component Placement

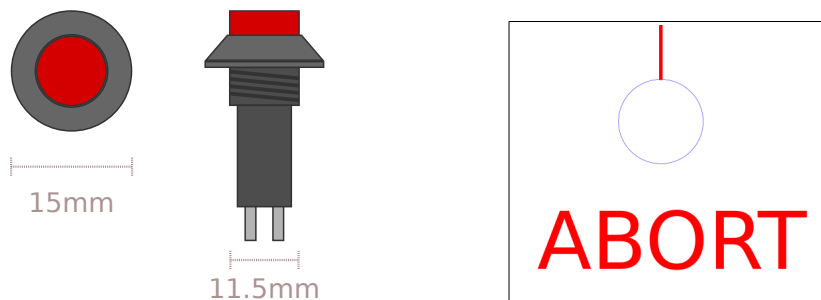
Most modules use just a handful of components, with a few specialty parts that are module- or function-specific.

Common buttons include the [Square Button](#), [Circle Button](#), [LED Button](#), and the [JH-D400X Joystick](#). In general, components have an orientation in which they should be inserted into the acrylic, which depends both on the function of the component as well as its pin/wire placement. These orientations are best determined by checking the PCBs which have dotted lines indicating where wires should connect, which should be matched by the location of the pins on the component.

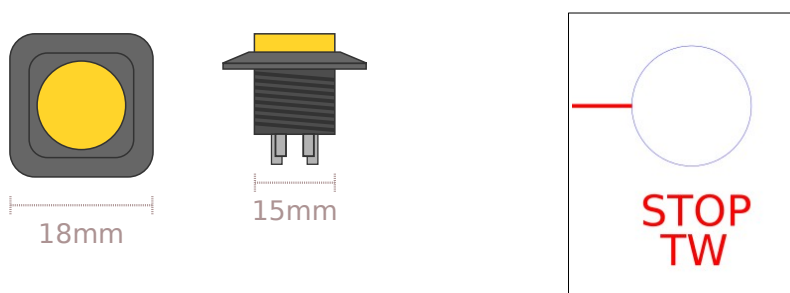
Square Buttons are about 14mm square on the surface, but underneath they are 11.85mm by 11mm. They fit into a hole 13mm x 11.2mm, with the extra space being taken up by the wings on either side. These wings compress to provide a stronger friction fit. However, they will typically only compress on one side, so it is worth making sure that compression happens on the same side uniform across your controller. The bottom side is preferred so the hole is better covered towards the user.



Circle Buttons are 15mm in diameter on the top and 11.5mm in diameter on the bottom. They fit into a hole 12mm in diameter.

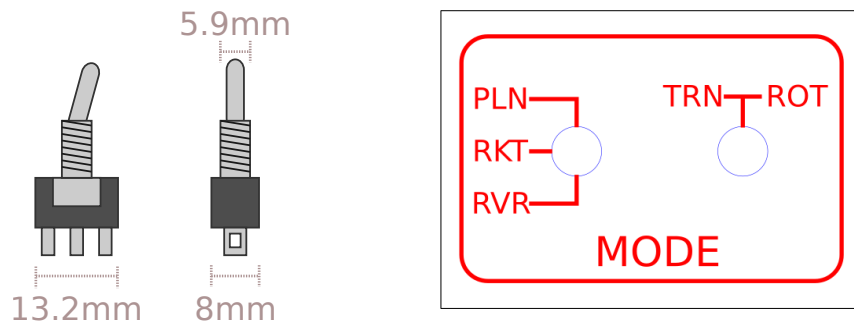


LED Buttons are 18mm in diameter on top and 15mm in diameter on the bottom. They have two sets of pins, one for the button and one for the LED. The pins for the LED are designated with + and – which should line up with the + and – on the PCB.

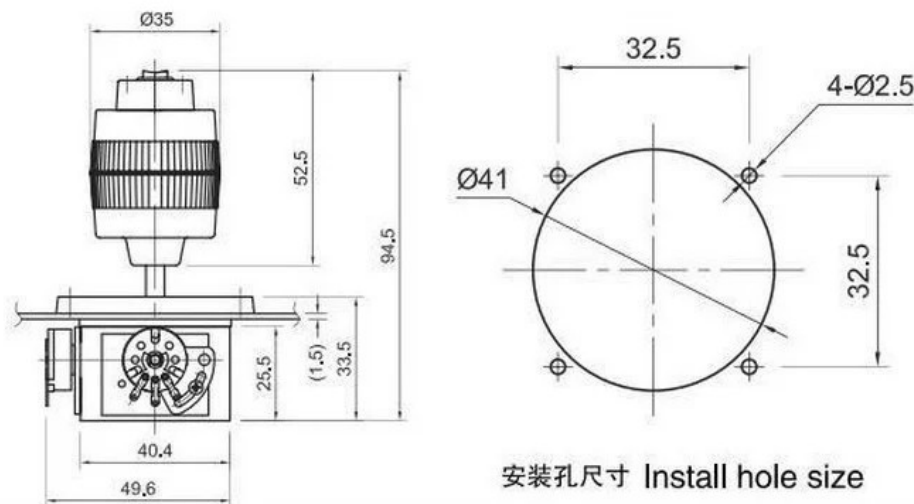


LED Switches are harder to source, but the ones used in production builds fit into a hole 20mm in diameter and have a similar 4-pin layout as the LED Buttons.

Switches come in a few varieties but they all share the same basic body plan, 13.2mm by 8mm at the base, with a threaded spire that is 5.9mm in diameter. They fit into a hole that is 6mm in diameter.

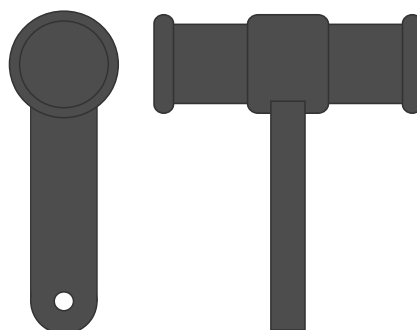


The JH-D400X Joystick is a complex part with 3 potentiometers and a pushbutton. Its dimensions are well-documented.



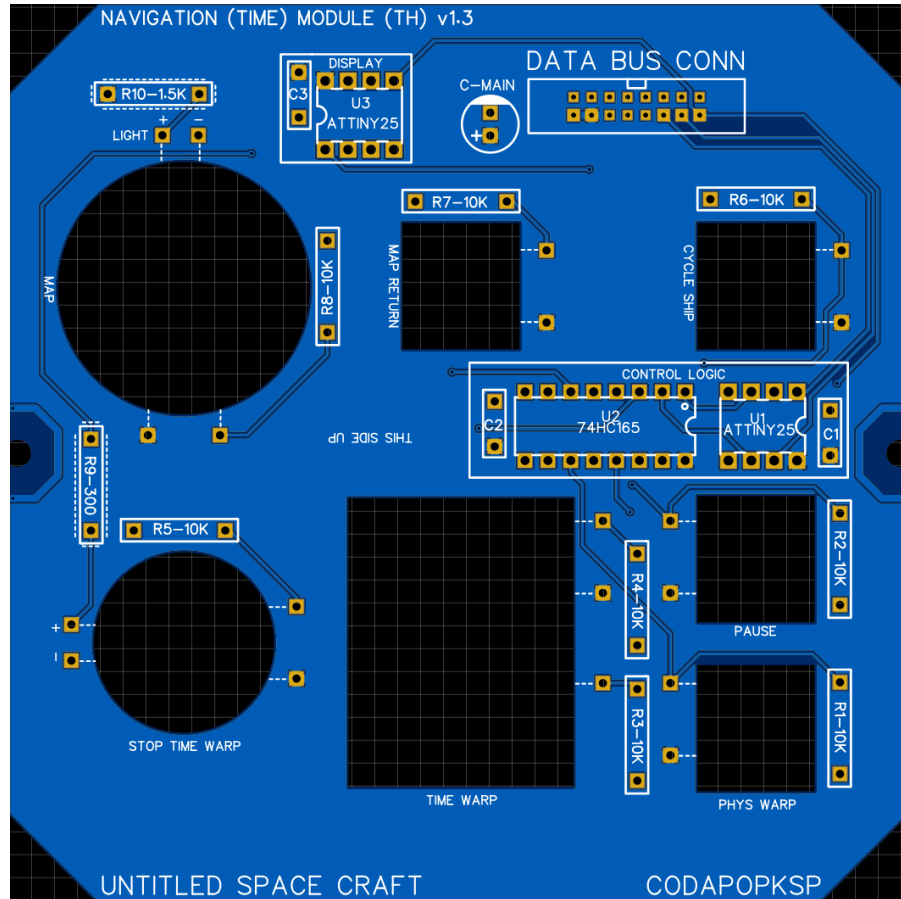
It has a rubber gasket that should go on top of the acrylic, with a ring to secure it. It's orientation can be determined by the shape of the hole in the PCBs. These joysticks often arrive from the manufacturer with poor calibration, so it is recommended to test the resistance of the stick at rest and its min/max in each axis before soldering.

The Throttle Lever is an STL file included in the USC repository. It includes a friction mechanism that snaps onto the bottom of the PCB. Both parts require more consideration than other components and will be covered in a later section. Their assembly occurs during or after soldering.

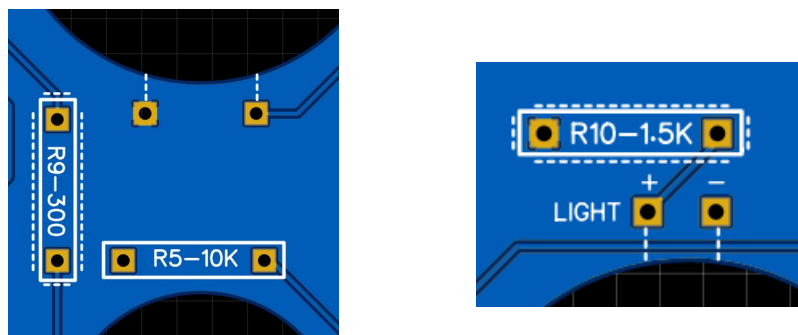


PCB Soldering

The PCBs for each module are filled with documentation which should be carefully read and considered before working. Every PCB in the project is single-sided, with the components and wires both being mounted on the side with documentation.



There are many resistors on each board, and over half of them are 10k Ohms. Resistors that aren't 10k Ohms are denoted with a dotted border. Some holes, particularly those near buttons or switches, have dotted lines pointing toward a component. These are to indicate the component they are a part of as well as where the component's pins should be for it to be in the proper orientation. The wire should go from its hole to the pin nearest to the dotted line.



Some holes are marked with a + or – symbol. These are used for LEDs or electrolytic capacitors that have a strict polarity. Some components also have + and – symbols on them. These should match the symbols on the PCB (+ to +). LEDs and electrolytic capacitors have longer or shorter legs to indicate + or – respectively.

Typically all PCB components such as resistors, capacitors, and chip sockets should be soldered and cleaned first, with the user interface components like buttons, switches, and LEDs added later.

The wires need to be cut and added to the PCB before they are connected to the user interface components. One way is to use a helping hands device to hold both the PCB and a wire and to solder it in through the bottom.



For LEDs to connect to the PCB, they often need to be bent in a U shape so their legs curve into the holes. Be sure not to let the legs touch each other. Below is an example.



This is the typical flow for most modules. Special cases will be discussed below.

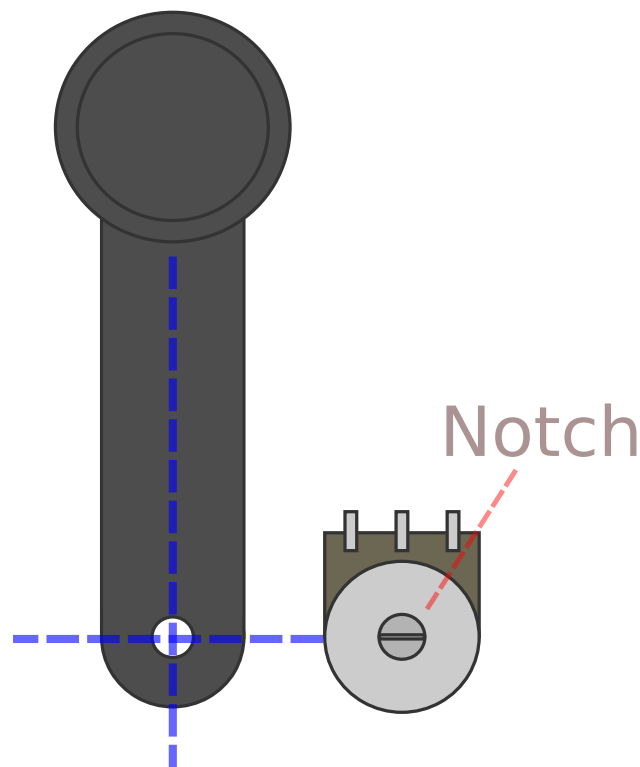
Throttle Module

Soldering the Throttle Module is more complex than most other modules and requires a very careful review of the sequence. At each point, think about how you are going to do the next several steps to ensure your plan makes sense.

Since the throttle lever needs to stick onto the end of the potentiometer, and the lever needs to fit through the holes of both the PCB and the acrylic, then this potentiometer cannot be soldered until later. Typically the sequence goes:

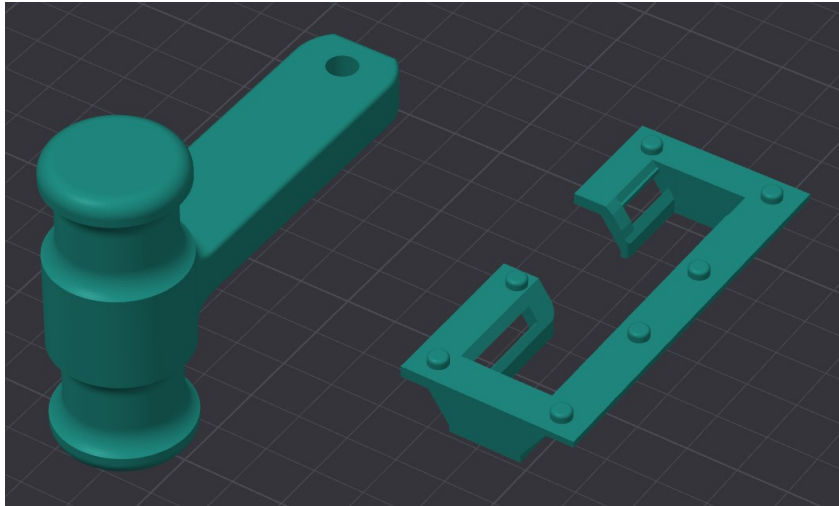
1. Solder all regular PCB components.
2. Solder all wires onto the PCB.
3. Stick all the buttons and switch onto the acrylic.
4. Feed the lever through the acrylic and PCB.
5. Fasten the lever onto the potentiometer.
6. Solder the potentiometer to the PCB.
7. Secure the PCB to the acrylic and solder the remaining wires to the components.

Extra care must also be taken to ensure the notch in the potentiometer is perpendicular to the shaft of the lever. This should be matched as closely as possible, though minor errors will be handled via calibration.



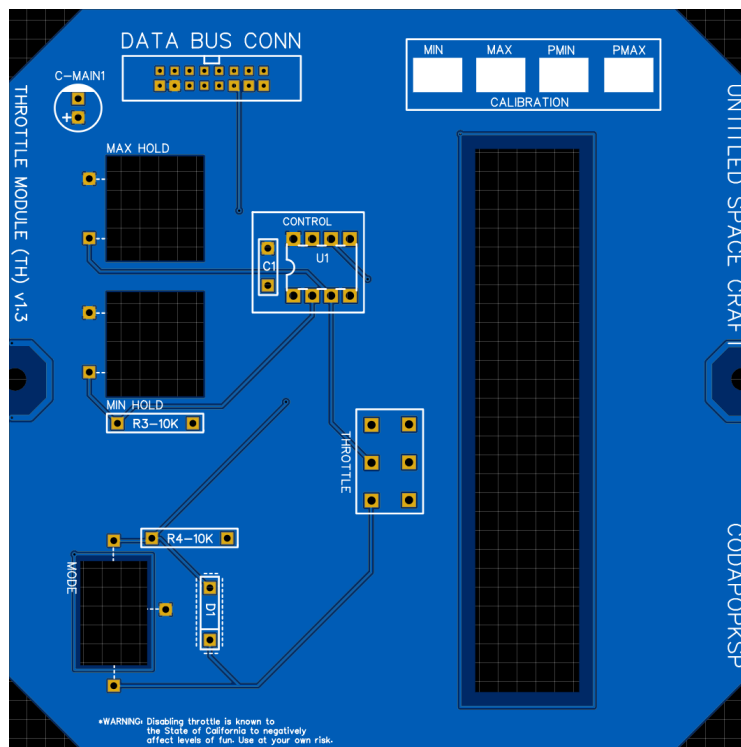
If printing with the unmodified STL files in the repository, the side facing the print bed should be the side that faces the potentiometer. This is because the opposite side is recessed a bit in order to fit the Throttle Guard.

The Throttle Guard is the name of the friction device that keeps the lever stationary when not actively being moved by the user. It fits into holes in the PCB and touches the lever underneath the module.



It is recommended to print several Throttle Guards and to treat each of them with different thickness of some other material wound around the contact bar (such as a thick tape like gaff tape). The number of winds of tape will alter the friction and compression of the lever. Too little, and the lever may move under gravity or inertia; too much and the Throttle Guard may plastically deform over time.

Once the fit is determined, the Throttle Guard should fit firmly into the PCB, but super glue can be used for added strength.



Calibration involves running the Throttle test script and reading the values in the Arduino Serial Monitor. First check the values of the throttle lever all the way down and all the way up, with the Mode switch in the ENBL (Enable) position. These values are the MIN and MAX. Then do the same with the Mode switch in PCSN (Precision). These are PMIN and PMAX.

It is recommended to write these values on the PCB in the designated areas. Then these values can be patched into the Settings.h file in the firmware prior to upload.

Telemetry Module

The Telemetry Module is the only module that uses an Arduino Nano rather than an ATTiny chip. This Arduino should be soldered directly to the PCB with the USB port aiming away from the center towards the edge.

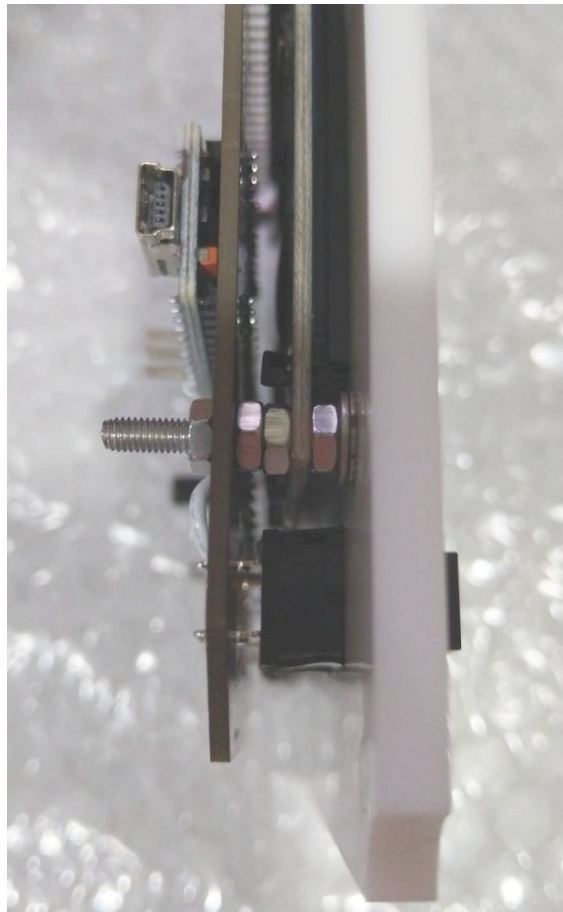
The screen itself needs to be carefully added, with washers and nuts used for spacing. The sequence goes:

1. Solder all regular PCB components.
2. Solder all wires onto the PCB.
3. Stick all the buttons onto the acrylic.
4. Feed the 4 long 3M screws through the PCB.
5. Add two thin 3M washers and a 3M nut to each screw for spacing.
6. Add the screen onto the screws.
7. Use two more 3M nuts onto each screw.
8. Push the PCB onto the screws and all the way down onto the pins for the screen.
9. Add the final nut to each screw and secure it tightly.
8. Solder the screen pins and button wires.

The final sequence should go:

[Nut] [PCB] [Nut] [Nut] [Screen PCB] [Nut] [Washer] [Washer] [Acrylic] [Screw Head]

If your washers are thicker, then you may only need one.



Joysticks

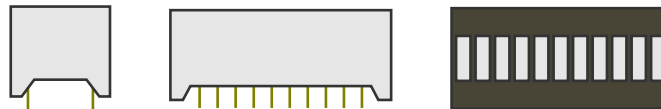
Each joystick has 6 pins sticking out the sides (from the potentiometers) and should be fairly straightforward to wire up, as they just connect to the PCB with jumpers from Axis 1 and Axis 2.

There are also 5 wires sticking out the side of the joystick: red, black, white, and two yellow. These go in the remaining empty holes next to the joystick labeled "Button" and "Axis 3". These holes are also labeled with letters that match the name of the color of the correct wire. R for red, B for black, and so on. I typically do this step after soldering the other wires to the PCB but before securing the PCB to the acrylic.



EVA Module

The EVA module is, for the most part, the same as the standard modules. The only difference is the 10 LED array for displaying EVA propellant. This array is organized with all of the anodes of the LEDs on one side (the side with the serial number sticker) and the cathodes on the other.

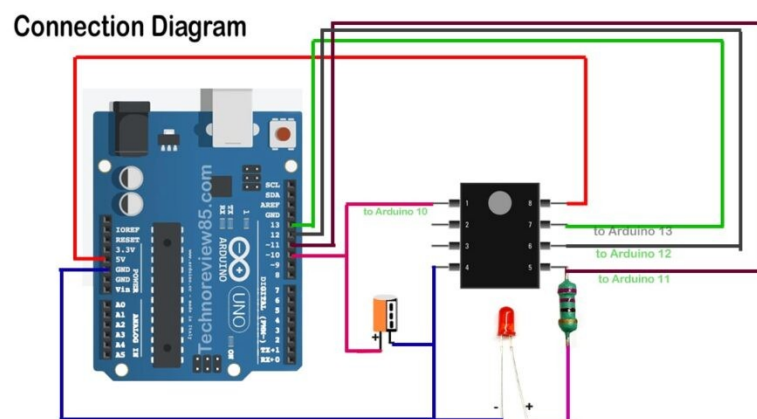


The array should be soldered to the PCB after the PCB and acrylic have already been joined, and it is nearly the final step of building the module. Soldering it first might prevent it from fitting into the hole in the acrylic later. The resulting array should stick up out of the acrylic slightly.

Adding the IC Chips

The ICs come in four different varieties. 74HC165, 74HC595, ATtinyX5, and ATtinyX4. It is recommended to use sockets for all your ICs to greatly reduce the time cost of fixing a module in event of a failure

The ATtinyX5 and ATtinyX4 chips are microcontrollers and need to be programmed. You can do this on a breadboard with directions for a simple ISP programmer circuit with [this guide](#).



Container Assembly

Untitled Space Craft containers contain module slots for holding modules, utilizing the same magnets that can be found in the modules. These magnets should always have the NORTH side UP in order to properly attract the magnets in the modules.

Wiring

Constructing the cables is a very crucial step with very little room for error. Your ribbon cables should have a red line along it that you can use for orientation. First you should make sure that your ribbon cable is 16 wires rather than 20. If it's 20, then you will need to remove 4 from the non-red side.

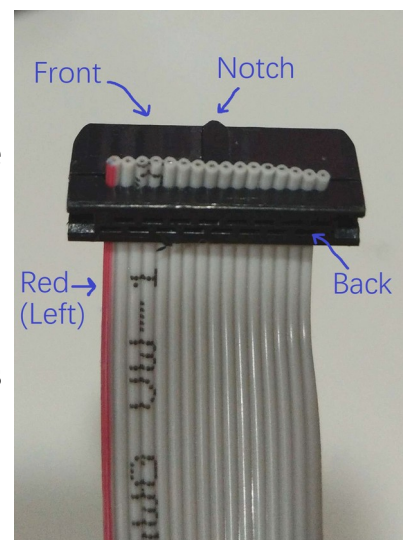


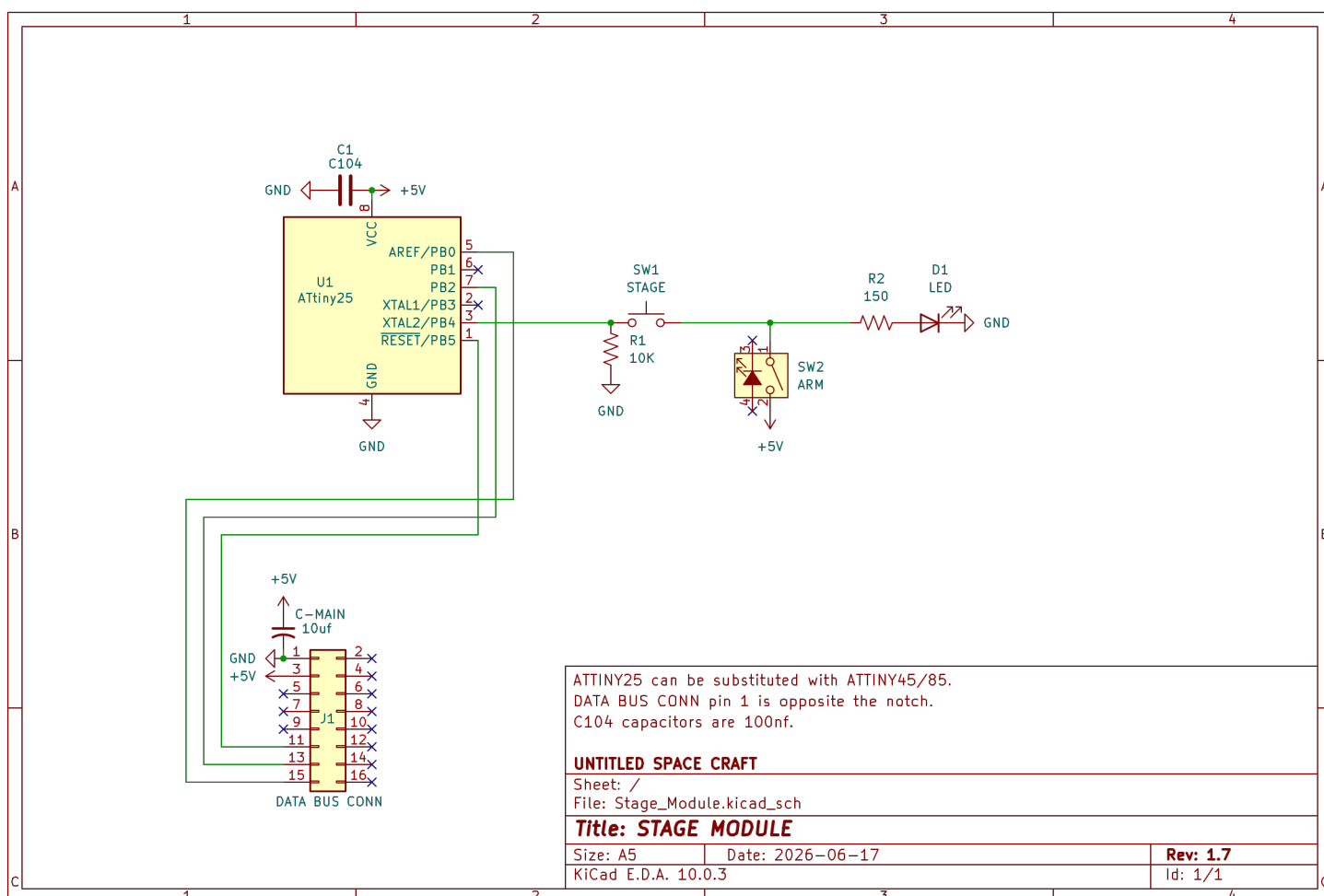
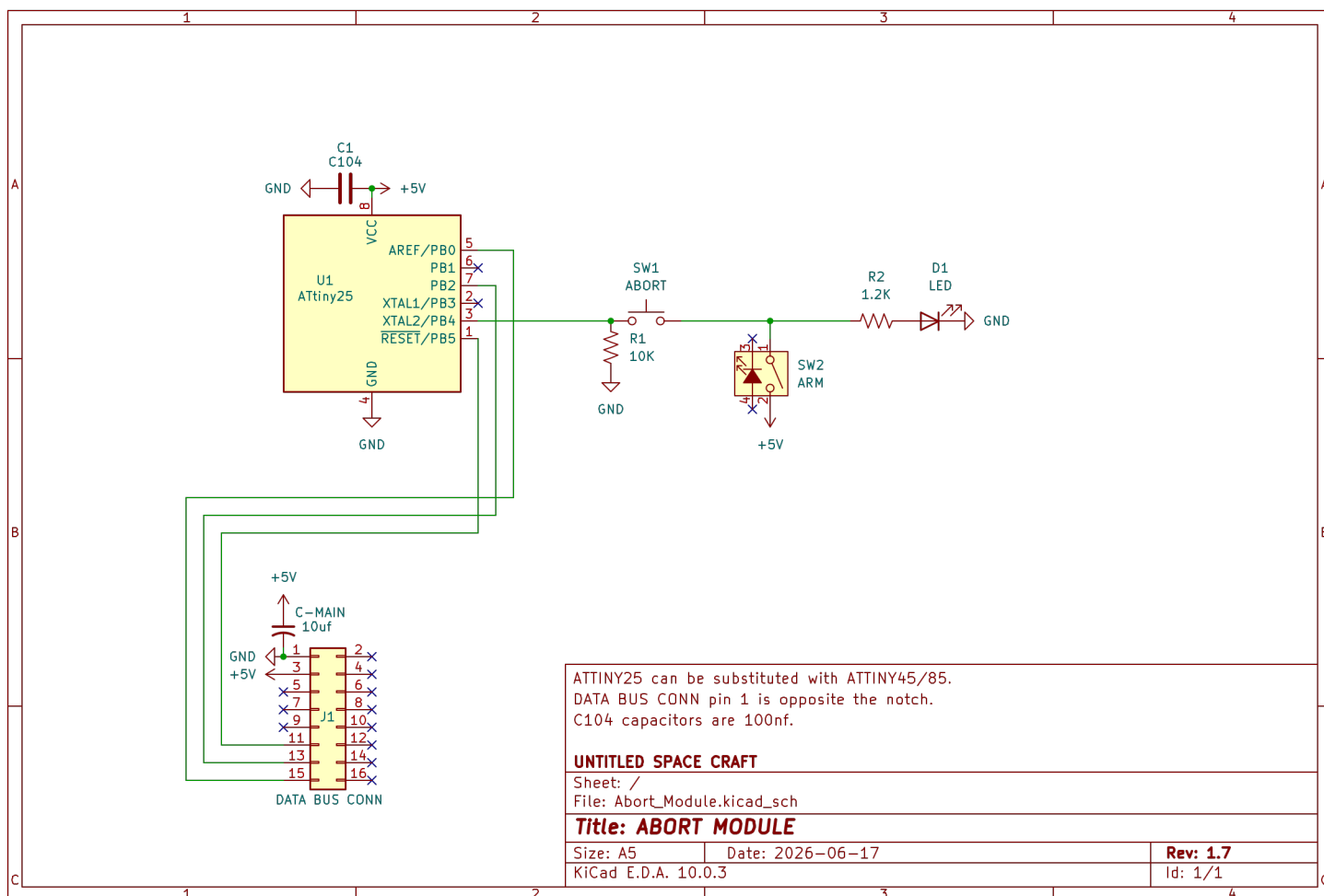
After that you should cut the wire into segments to fit your container. This can be done by counting how many modules removed a particular module is away from the hub. The module that sits on top of the hub is 1. Every module that is adjacent to that module is 2. And every module that is adjacent to a 2 is a 3, and so on. Determining that, you should cut the wires as so:

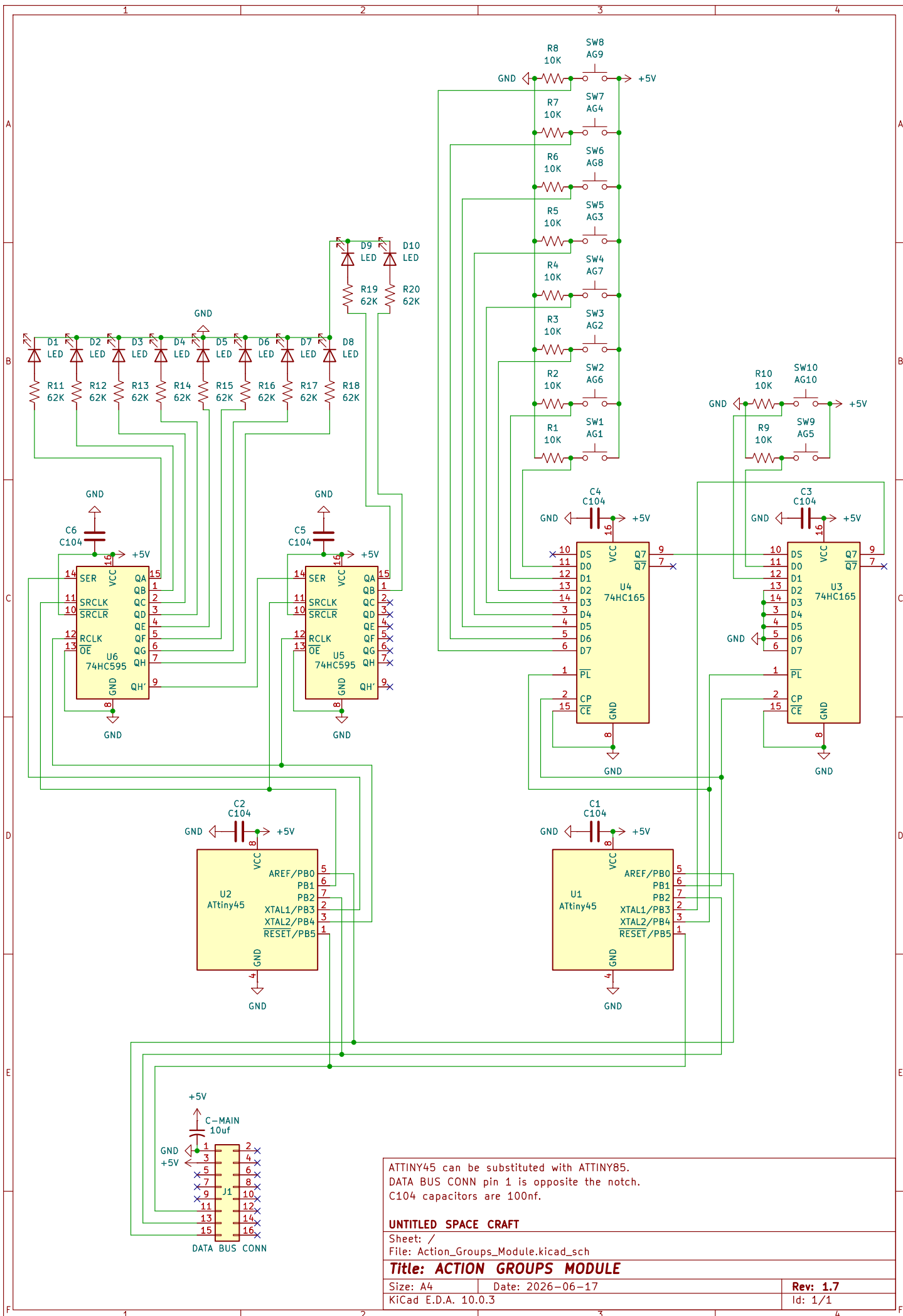
- 1: 12cm
- 2: 24cm
- 3: 36cm
- 4: 48cm

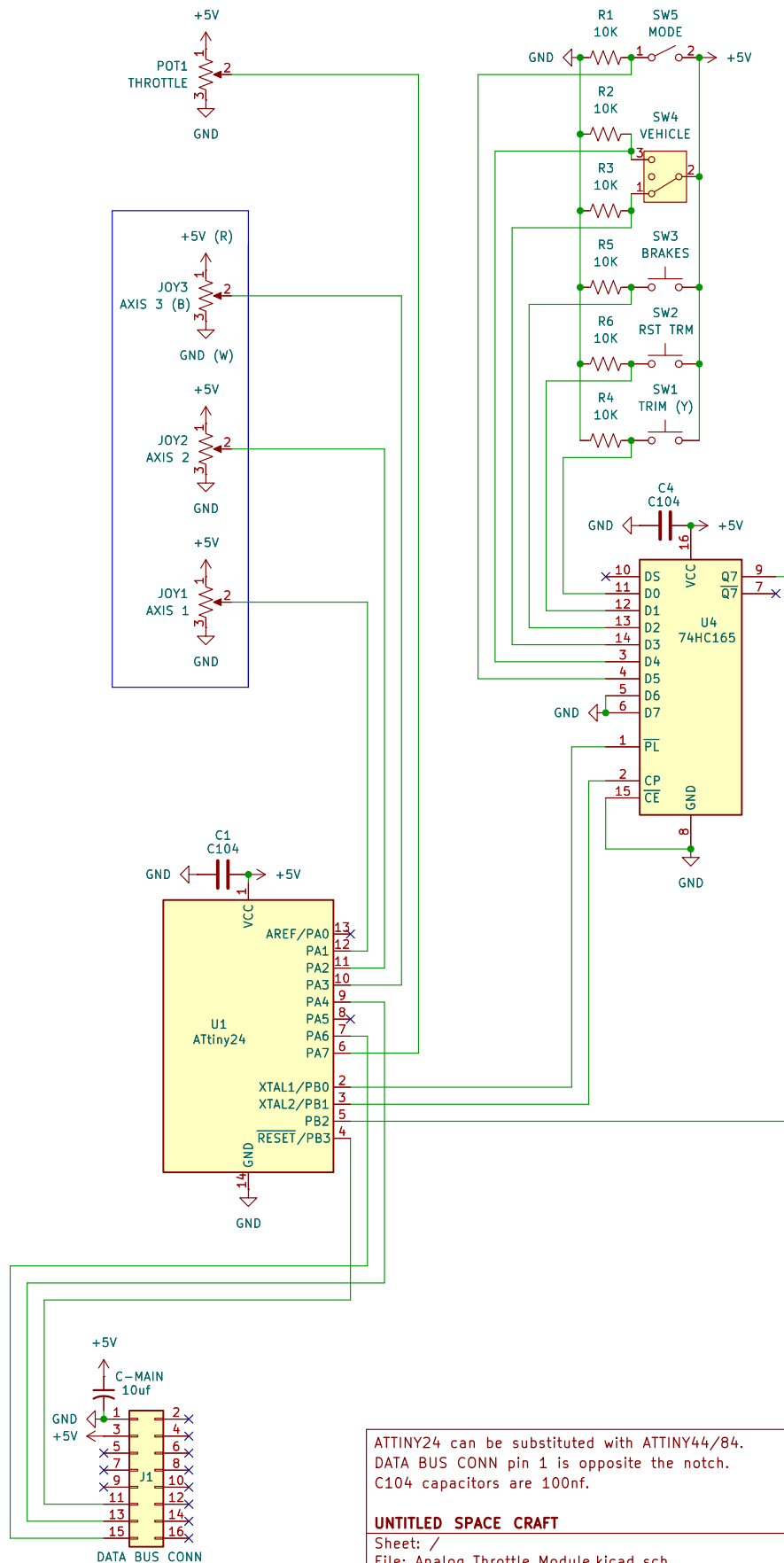
If you hold the wire in front of you with the red line towards the left, then the connector should be facing away from you and with the notch facing up.

Insert the cable into the connector and then press the connector together in the vise, causing it to combine with the cable. Every connector should be added in the same way, with the connectors always facing the same way in relation to the red line. This ensures that the wires can be used in either orientation and interchangeably. If any wires are crossed here, it can be disastrous for your ICs.









ATTINY24 can be substituted with ATTINY44/84.
DATA BUS CONN pin 1 is opposite the notch.
C104 capacitors are 100nf.

UNTITLED SPACE CRAFT

Sheet: /

File: Analog_Throttle_Module.kicad_sch

Title: ANALOG THROTTLE MODULE

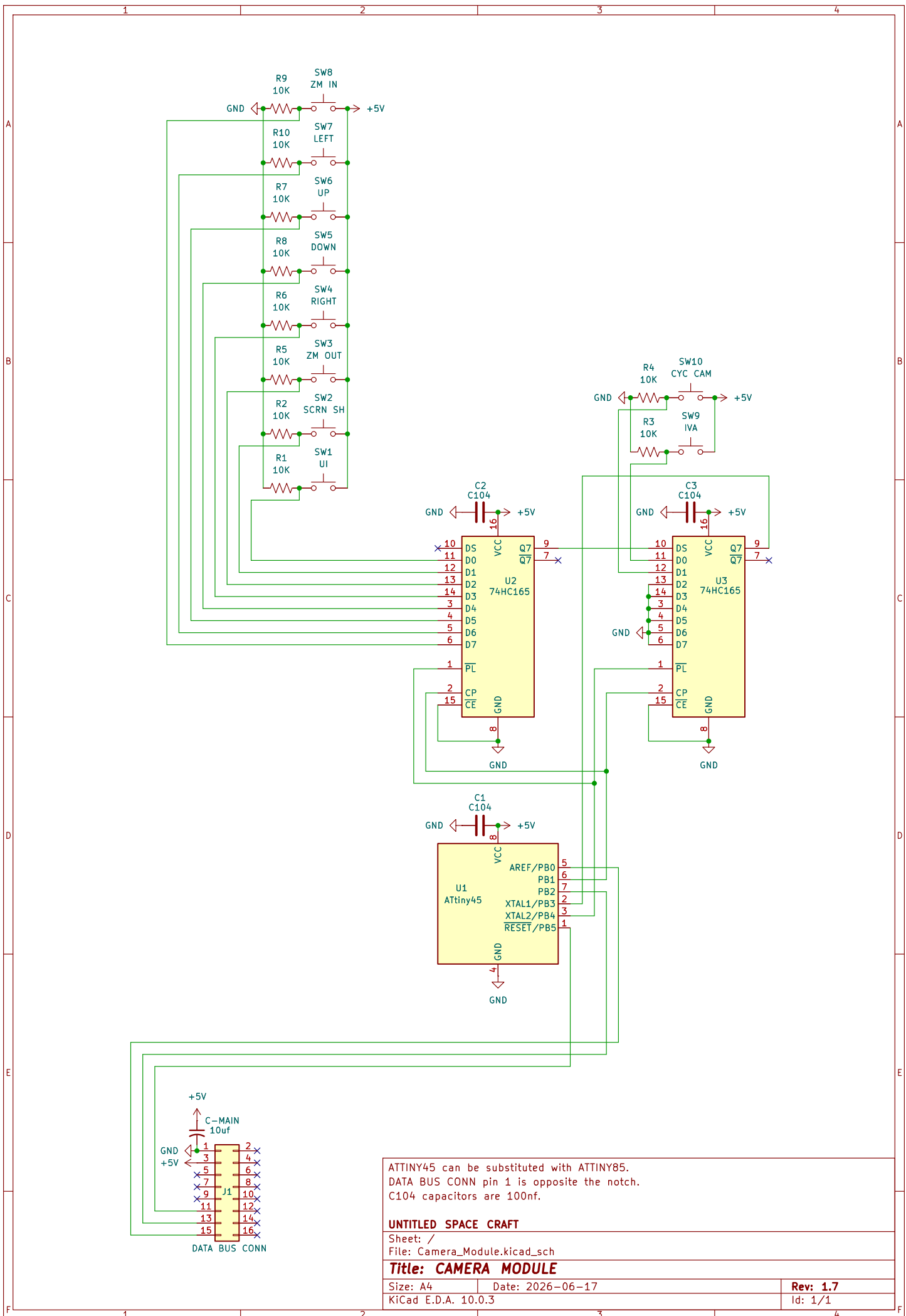
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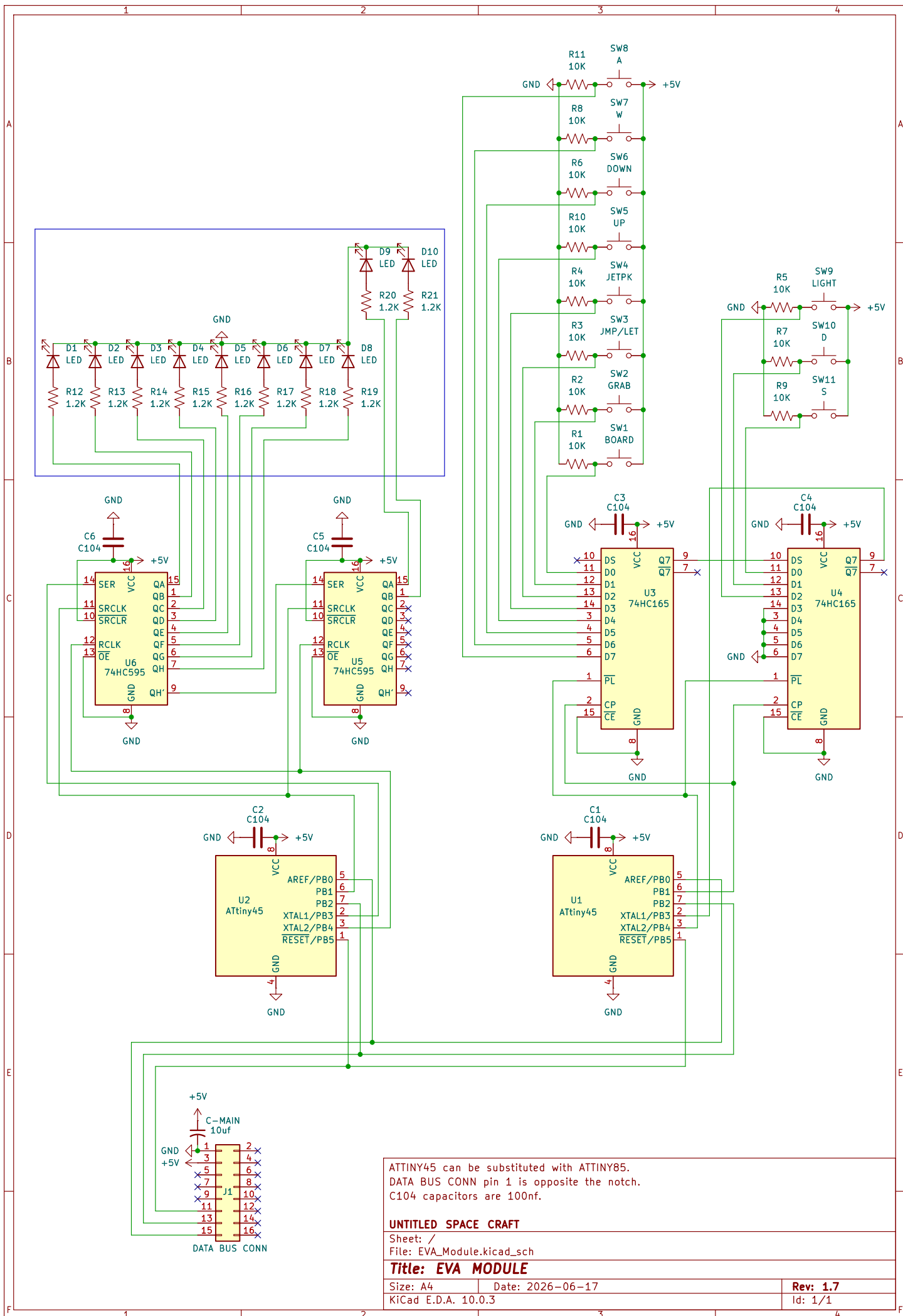
Date: 2026-06-17

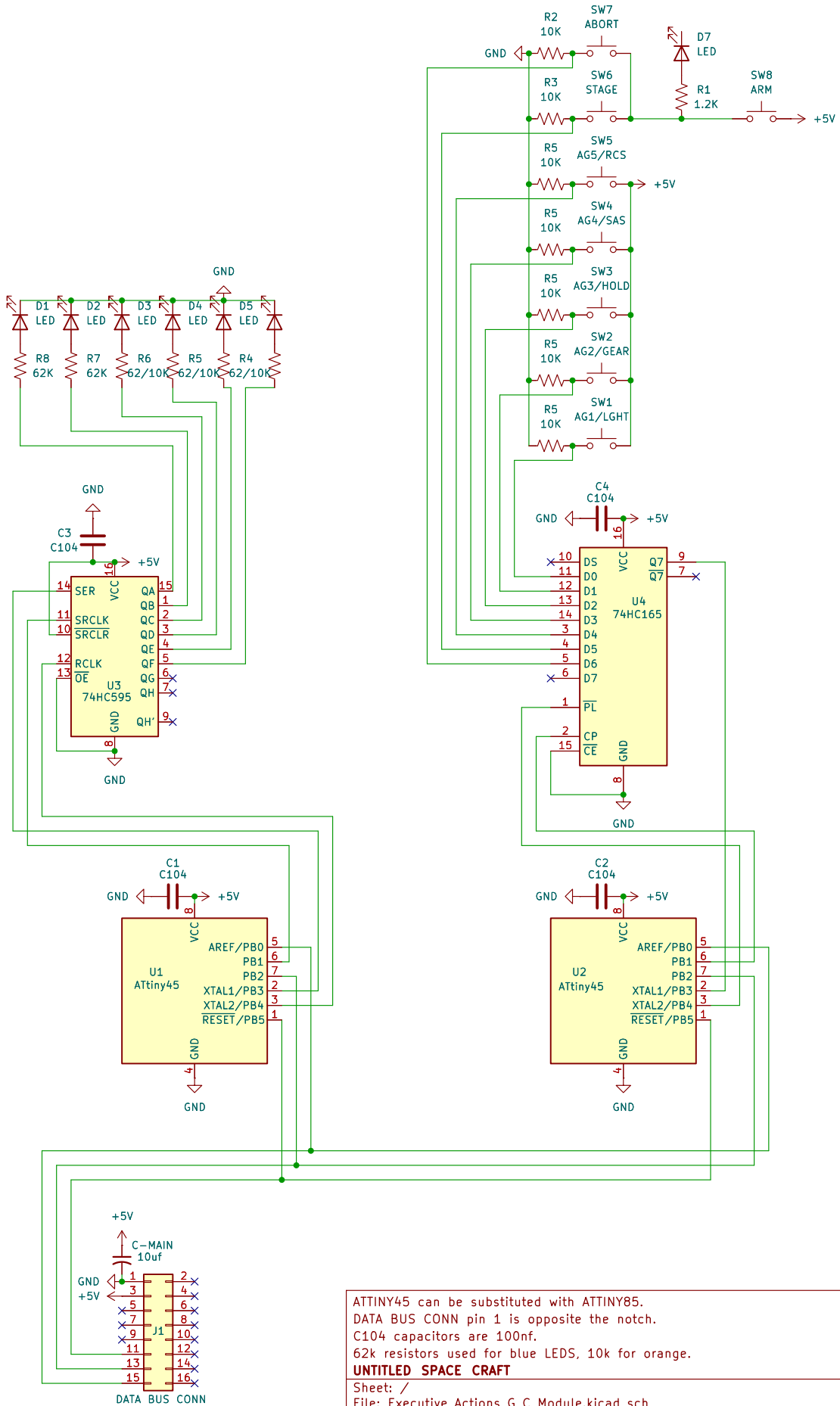
Rev: 1.7

KiCad E.D.A. 10.0.3

Id: 1/1







ATTINY45 can be substituted with ATTINY85.
 DATA BUS CONN pin 1 is opposite the notch.
 C104 capacitors are 100nf.
 62k resistors used for blue LEDS, 10k for orange.

UNTITLED SPACE CRAFT

Sheet: /

File: Executive_Actions_G_C_Module.kicad_sch

Title: EXECUTIVE ACTIONS (GROUPS/CONTROL) MODULE

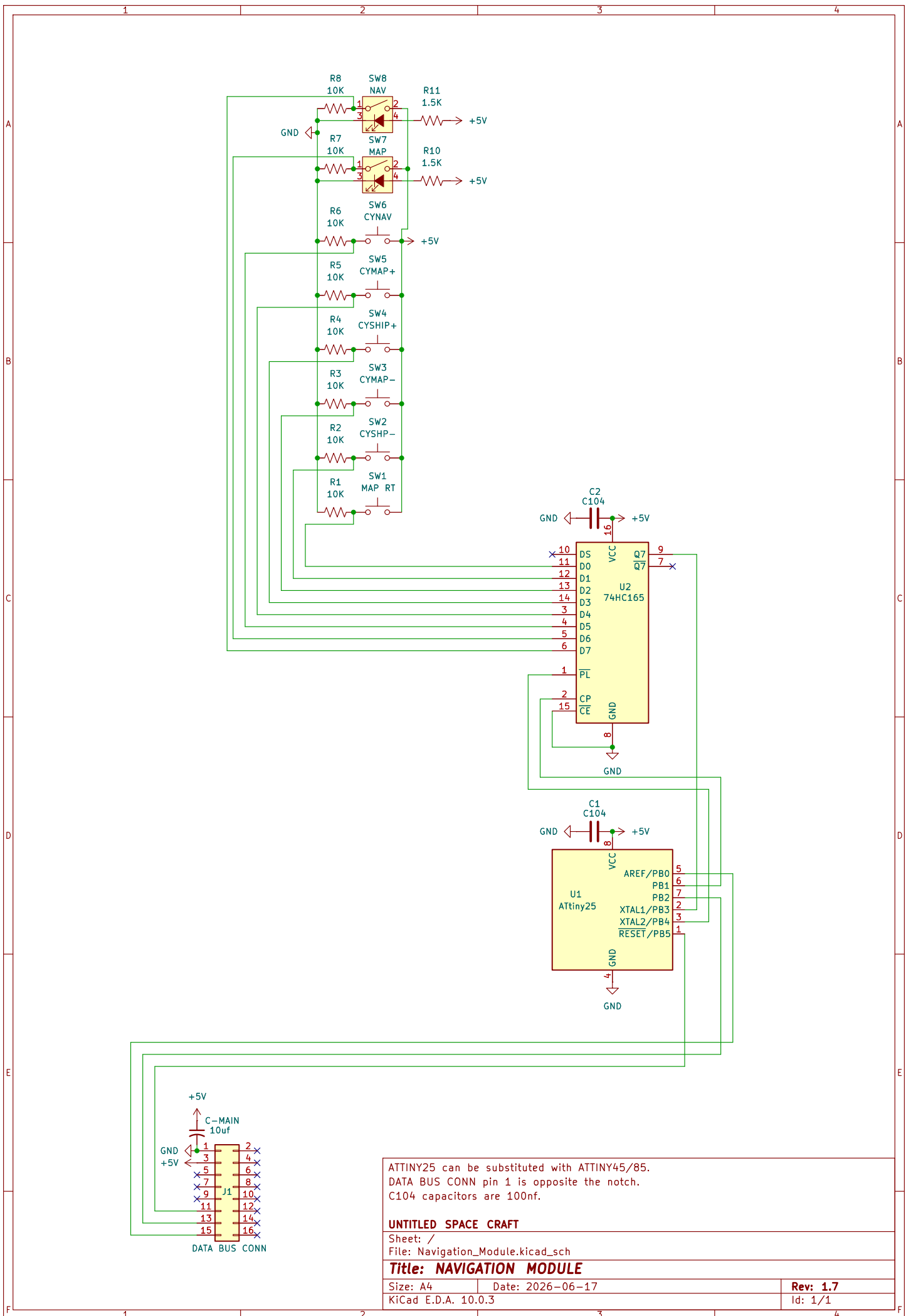
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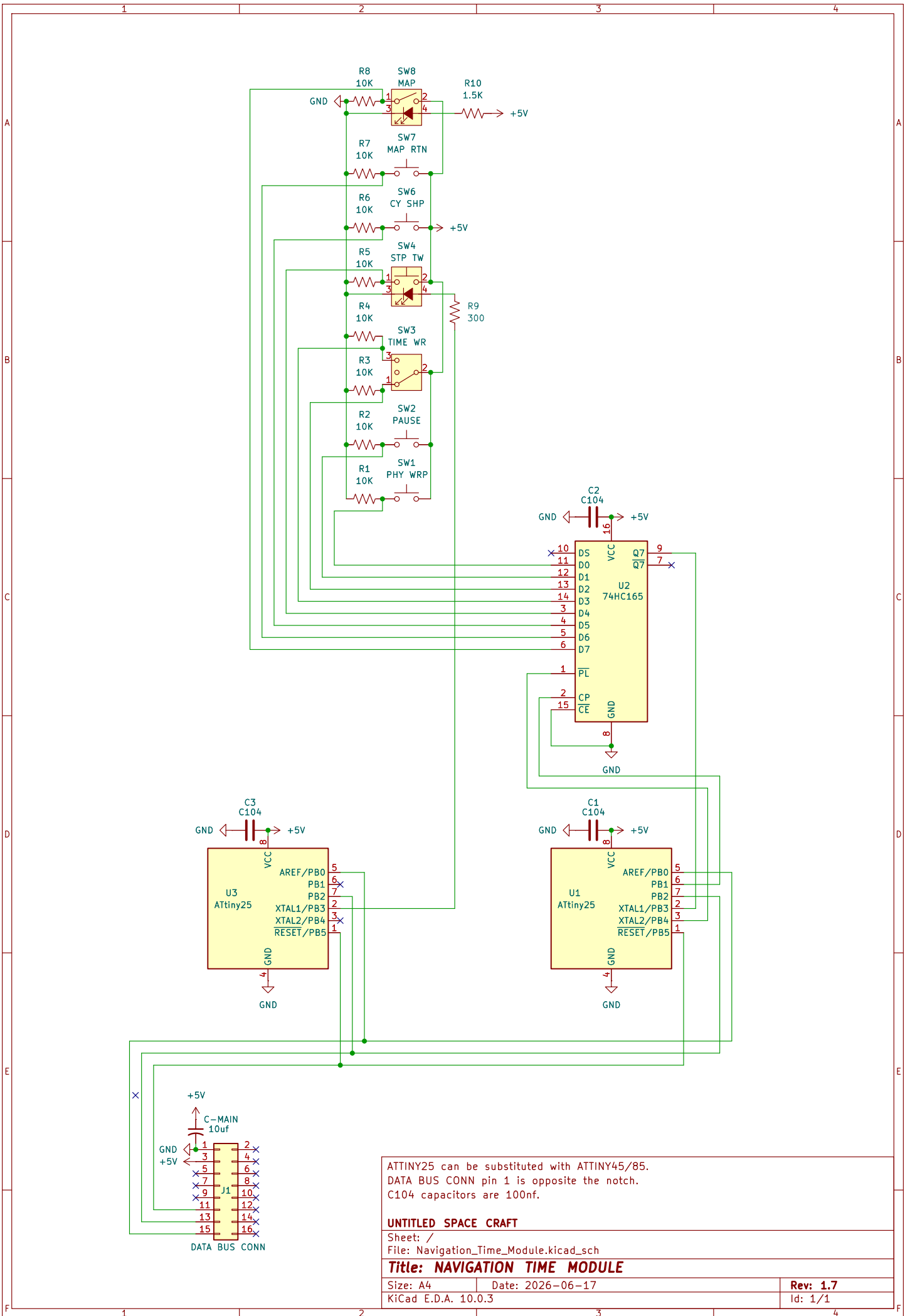
Date: 2026-06-17

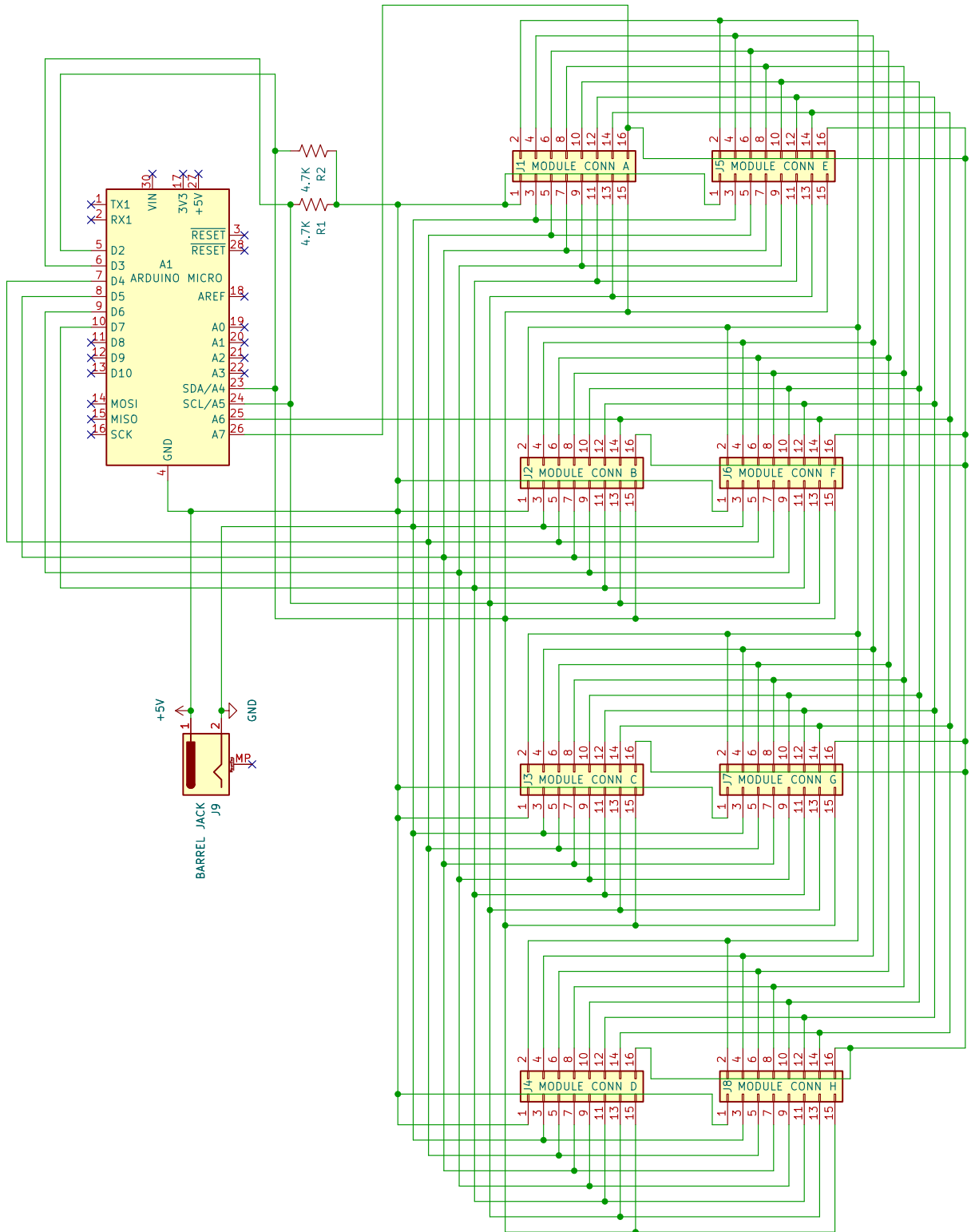
Rev: 1.6

KiCad E.D.A. 10.0.3

Id: 1/1







Pins are for Arduino Nano, but an Arduino Micro should fit into the same slot to enable keyboard emulation.
Barrel Jack: Tip/inside is 5V, Sleeve is GND.

UNTITLED SPACE CRAFT

Sheet: /

File: Octo_Hub.kicad_sch

Title: OCTO HUB

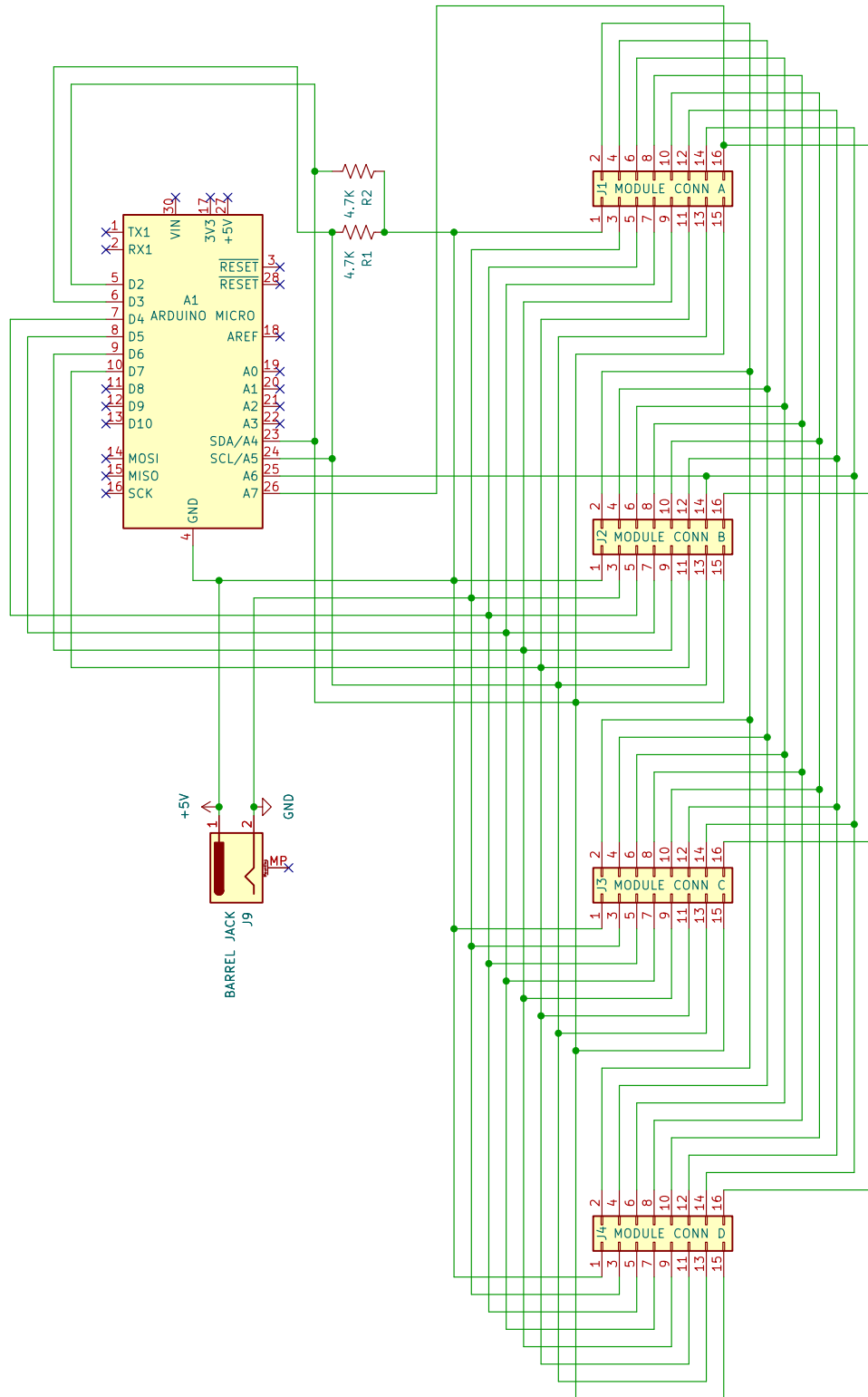
Size: A4

Date: 2026-06-17

KiCad E.D.A. 10.0.3

Rev: 1.7

Id: 1/1



Pins are for Arduino Nano, but an Arduino Micro should fit into the same slot to enable keyboard emulation.
Barrel Jack: Tip/inside is 5V, Sleeve is GND.

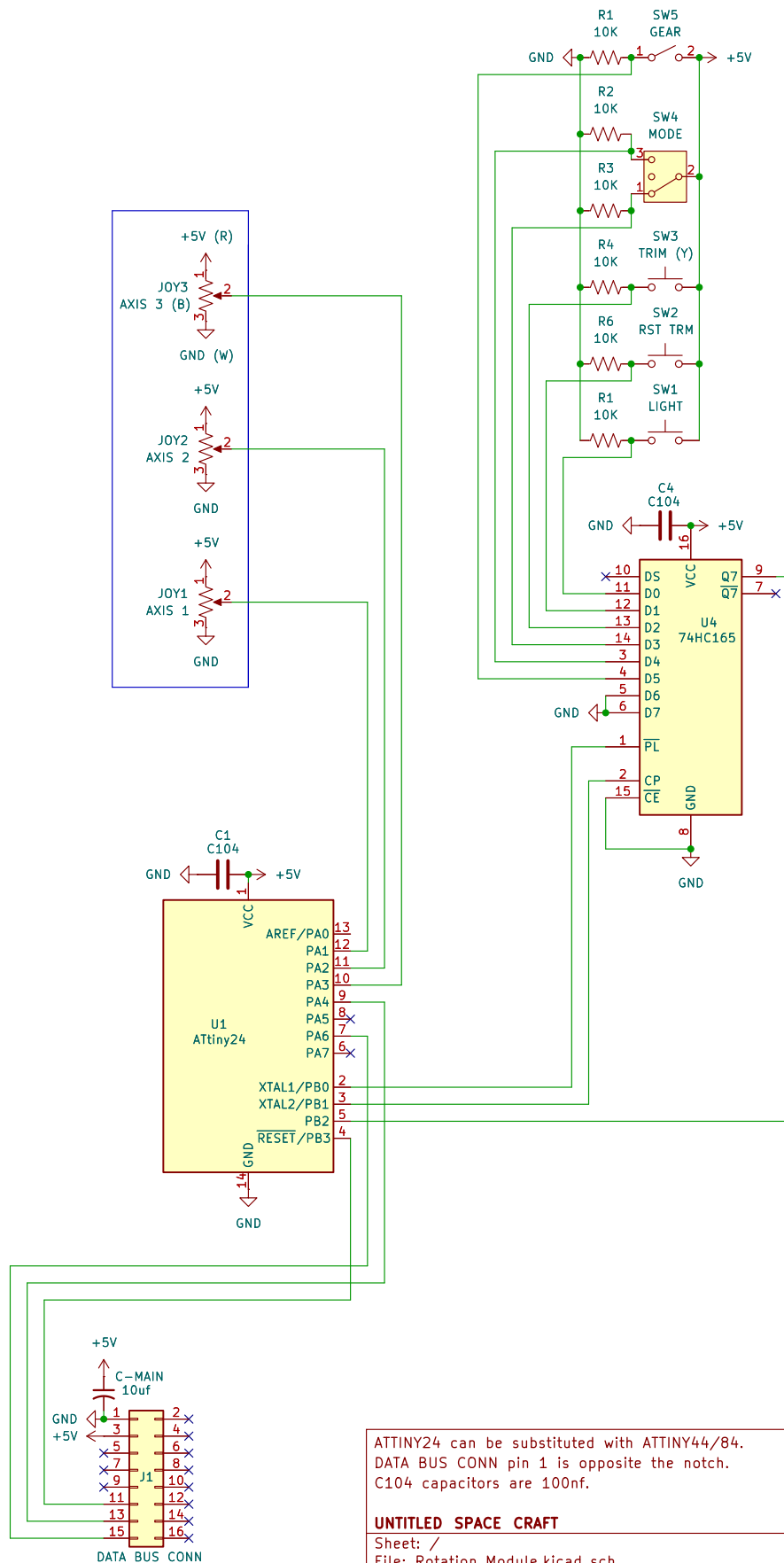
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Sheet: /
File: Quad_Hub.kicad_sch

Title: QUAD HUB

Size: A4 Date: 2026-06-17
KiCad E.D.A. 10.0.3

Rev: 1.7
Id: 1/1



ATTINY24 can be substituted with ATTINY44/84.
DATA BUS CONN pin 1 is opposite the notch.
C104 capacitors are 100nf.

UNTITLED SPACE CRAFT

Sheet: /

File: Rotation_Module.kicad_sch

Title: ROTATION MODULE

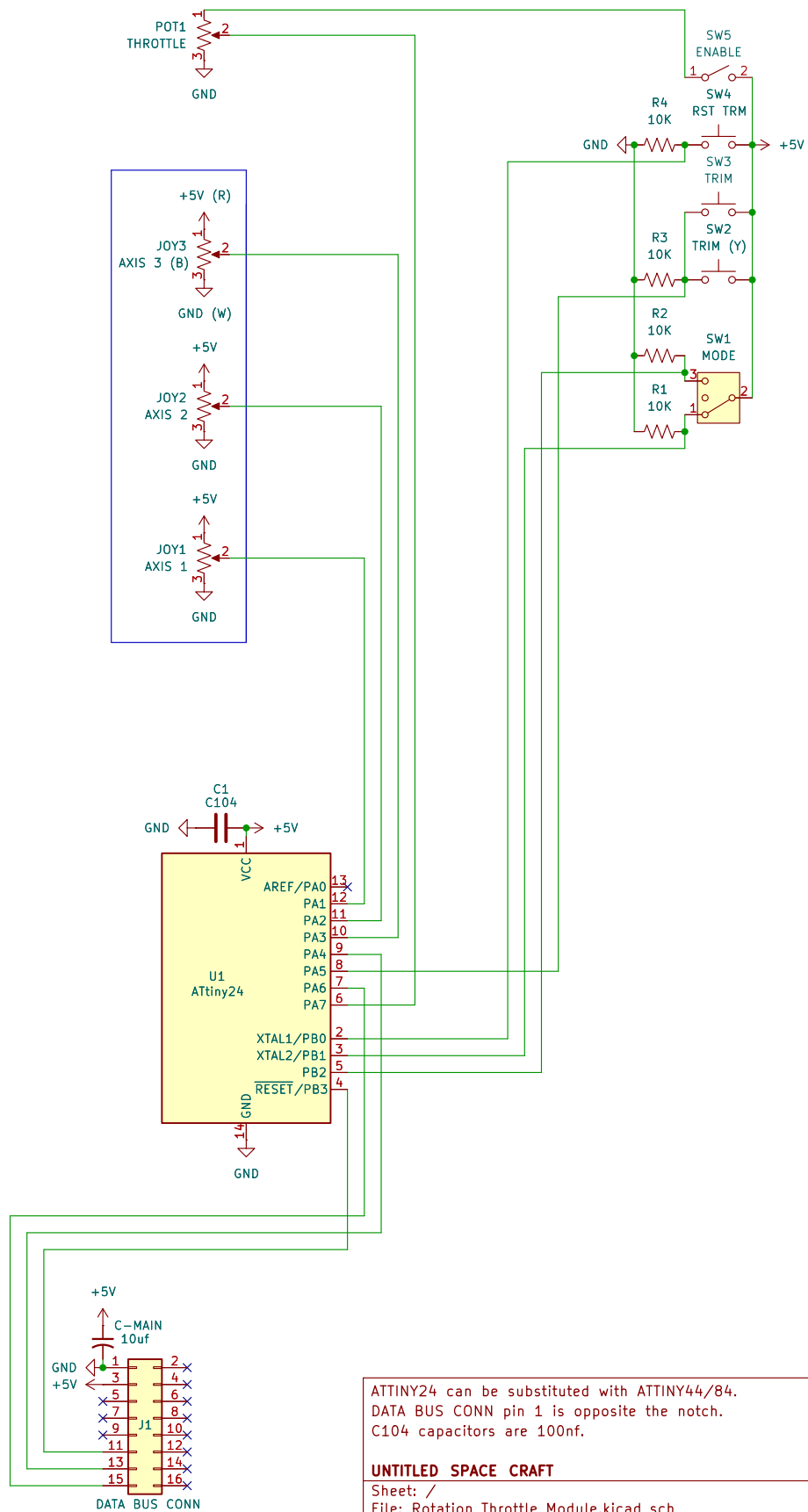
Size: A4

Date: 2026-06-17

Rev: 1.7

KiCad E.D.A. 10.0.3

Id: 1/1



ATTINY24 can be substituted with ATTINY44/84.
DATA BUS CONN pin 1 is opposite the notch.
C104 capacitors are 100nf.

UNTITLED SPACE CRAFT

Sheet: /

File: Rotation_Throttle_Module.kicad_sch

Title: ANALOG THROTTLE MODULE

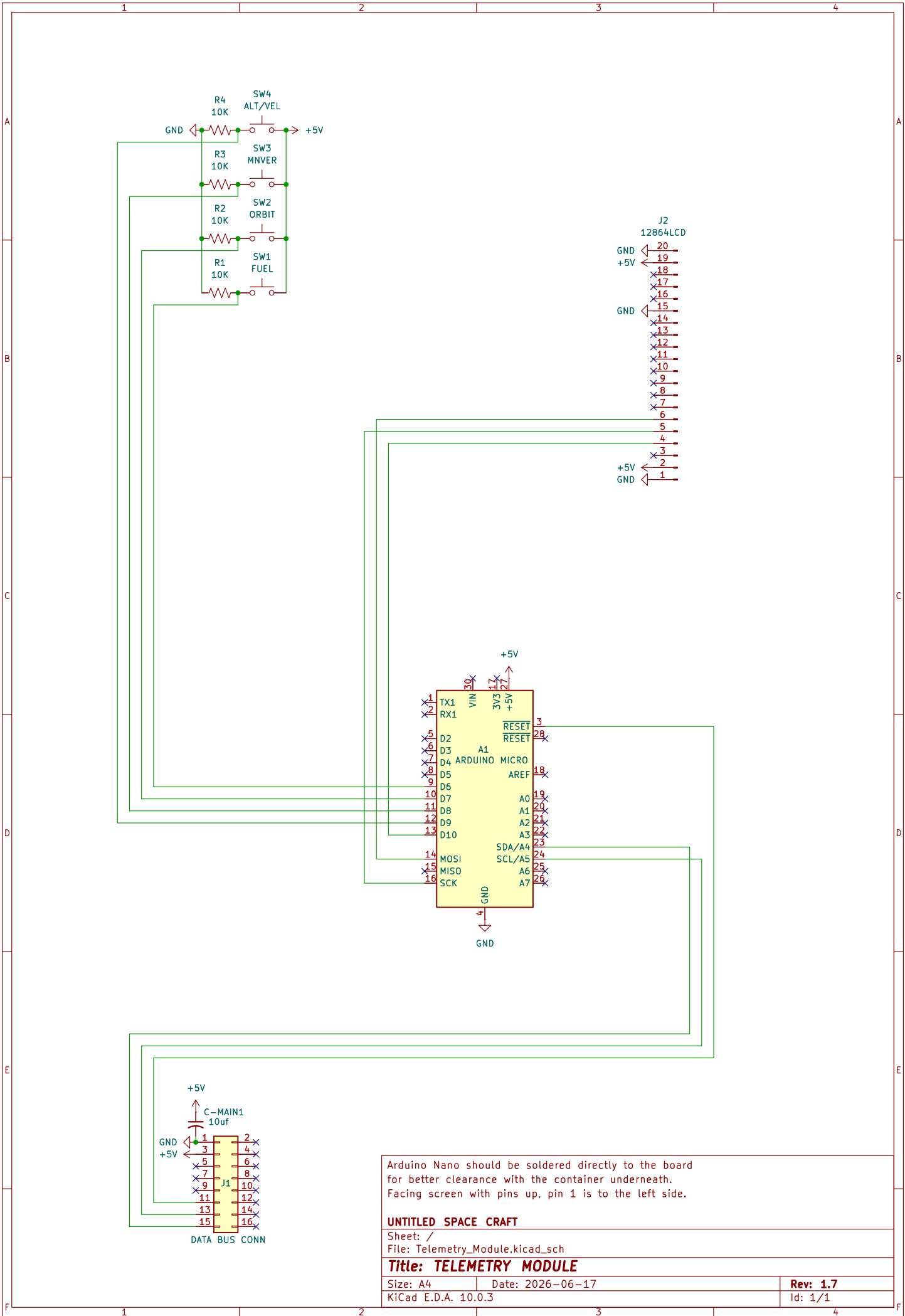
Size: A4

Date: 2026-06-17

Rev: 1.7

KiCad E.D.A. 10.0.3

Id: 1/1



Arduino Nano should be soldered directly to the board for better clearance with the container underneath. Facing screen with pins up, pin 1 is to the left side.

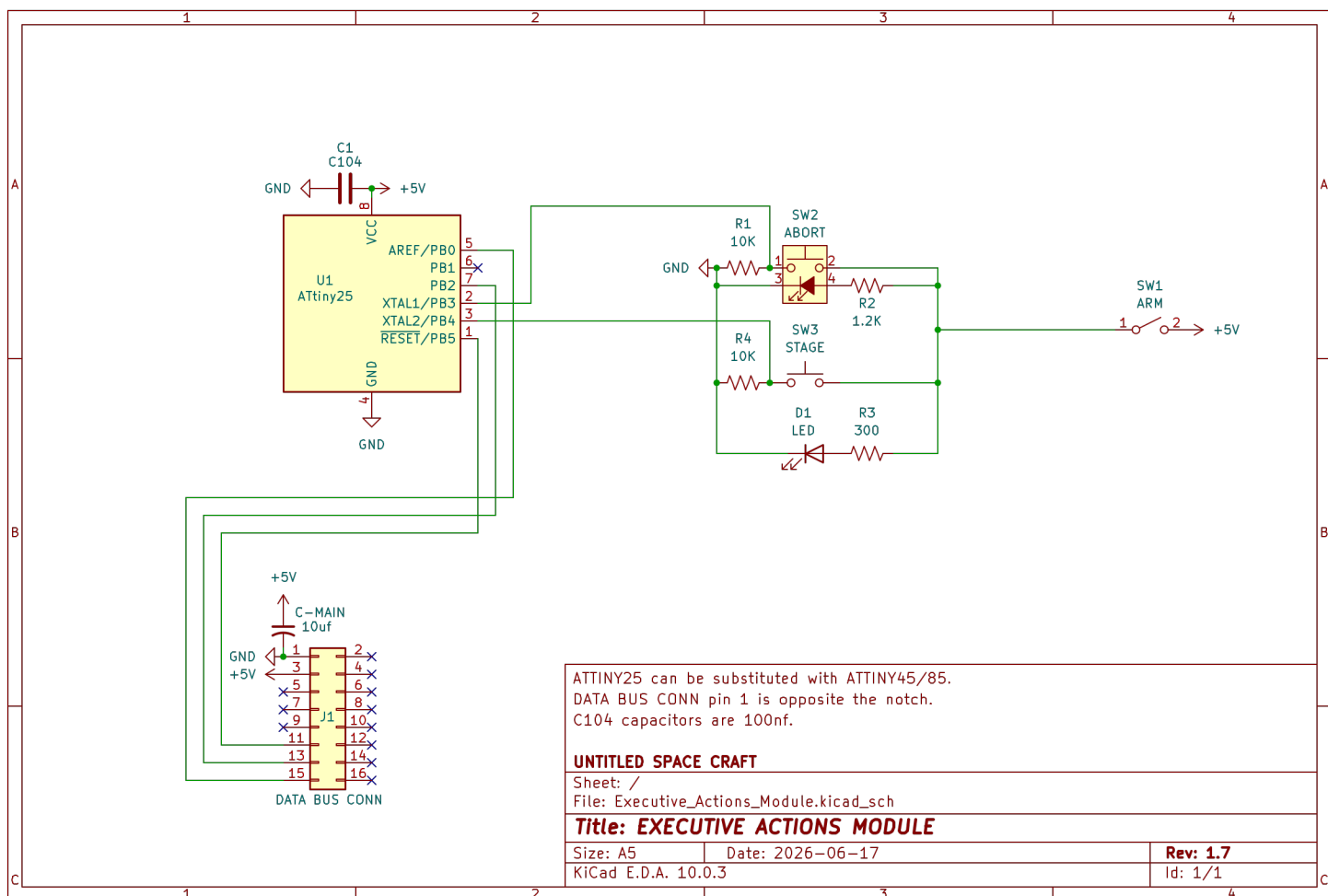
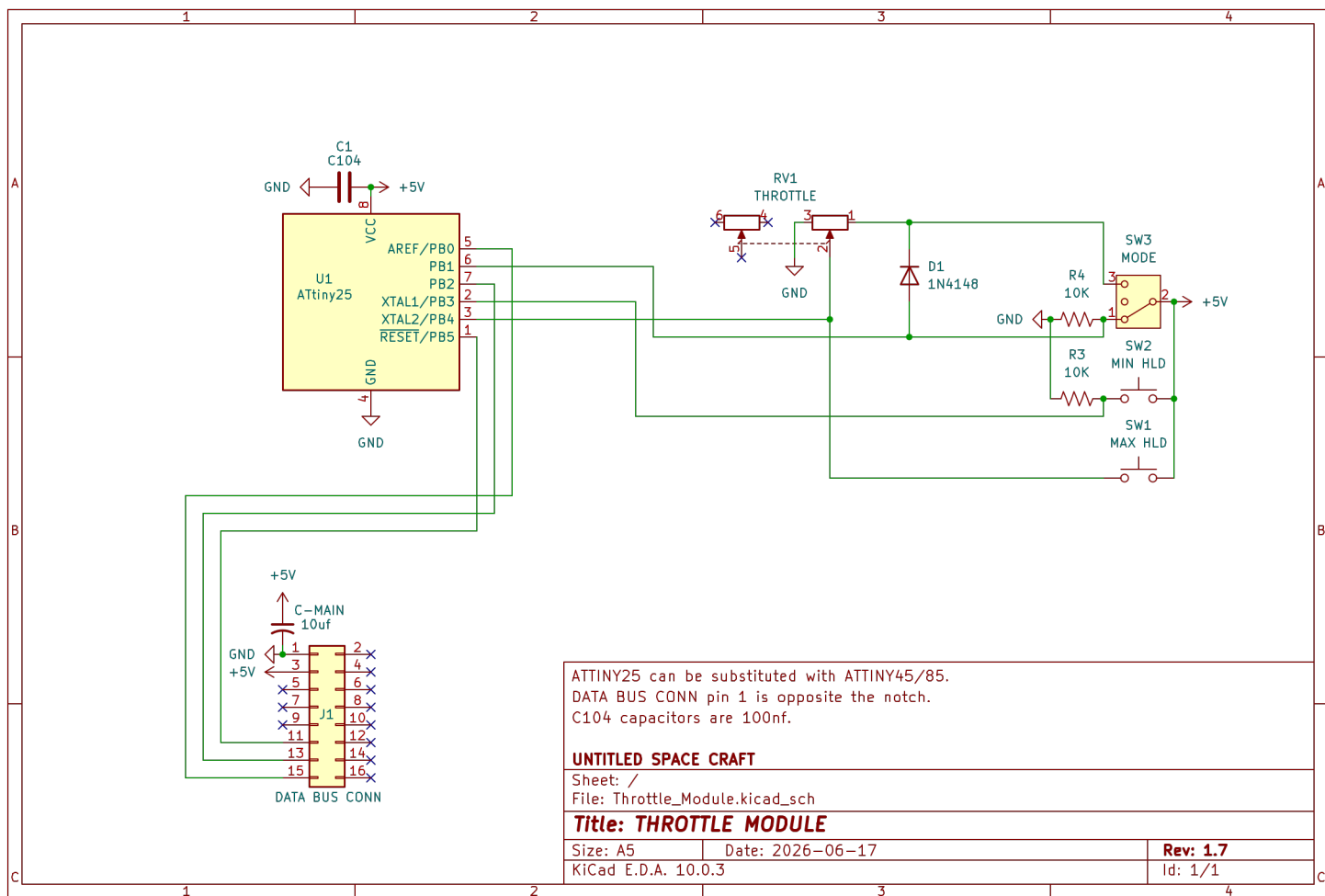
UNTITLED SPACE CRAFT

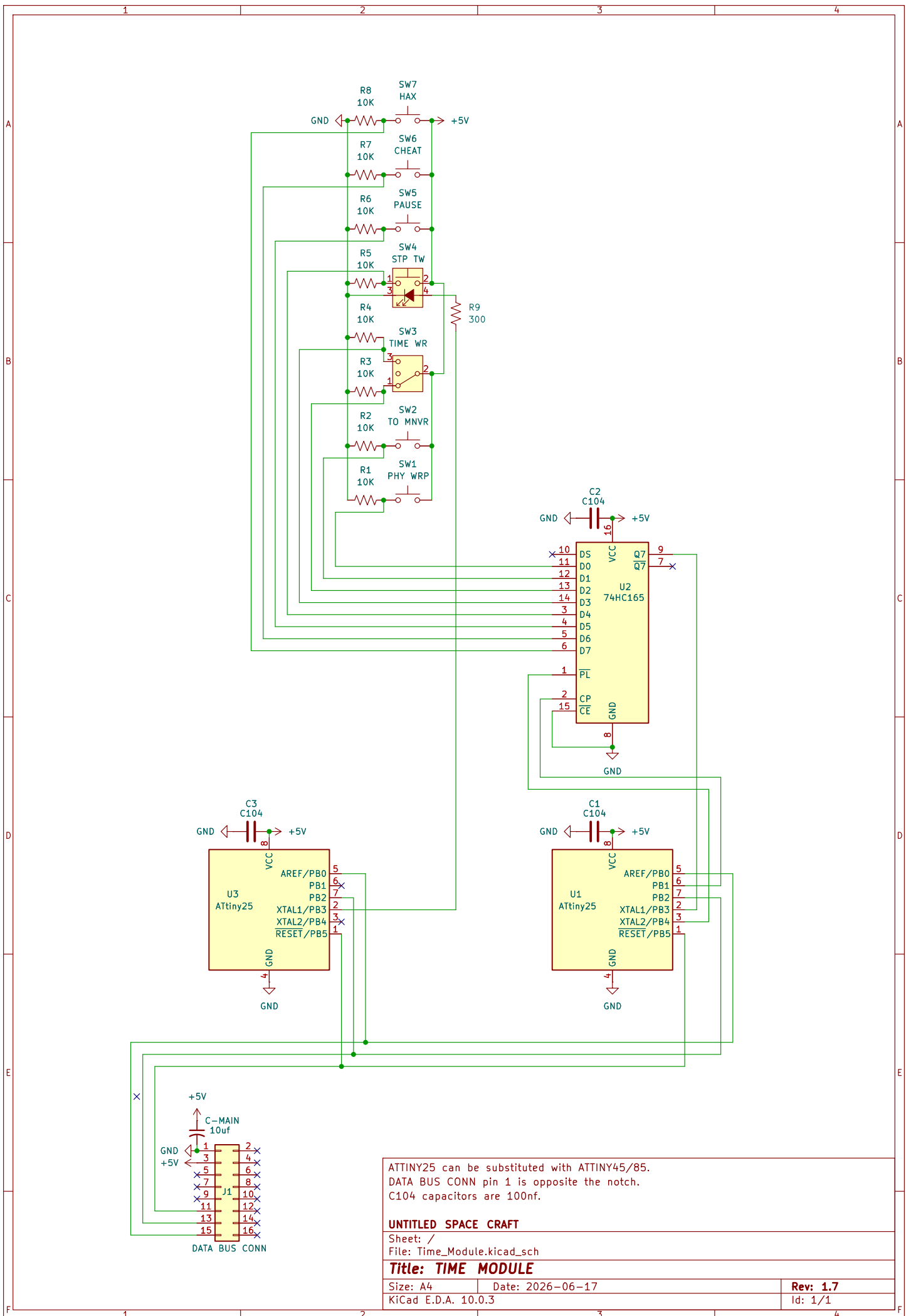
Sheet: /
File: Telemetry_Module.kicad_sch

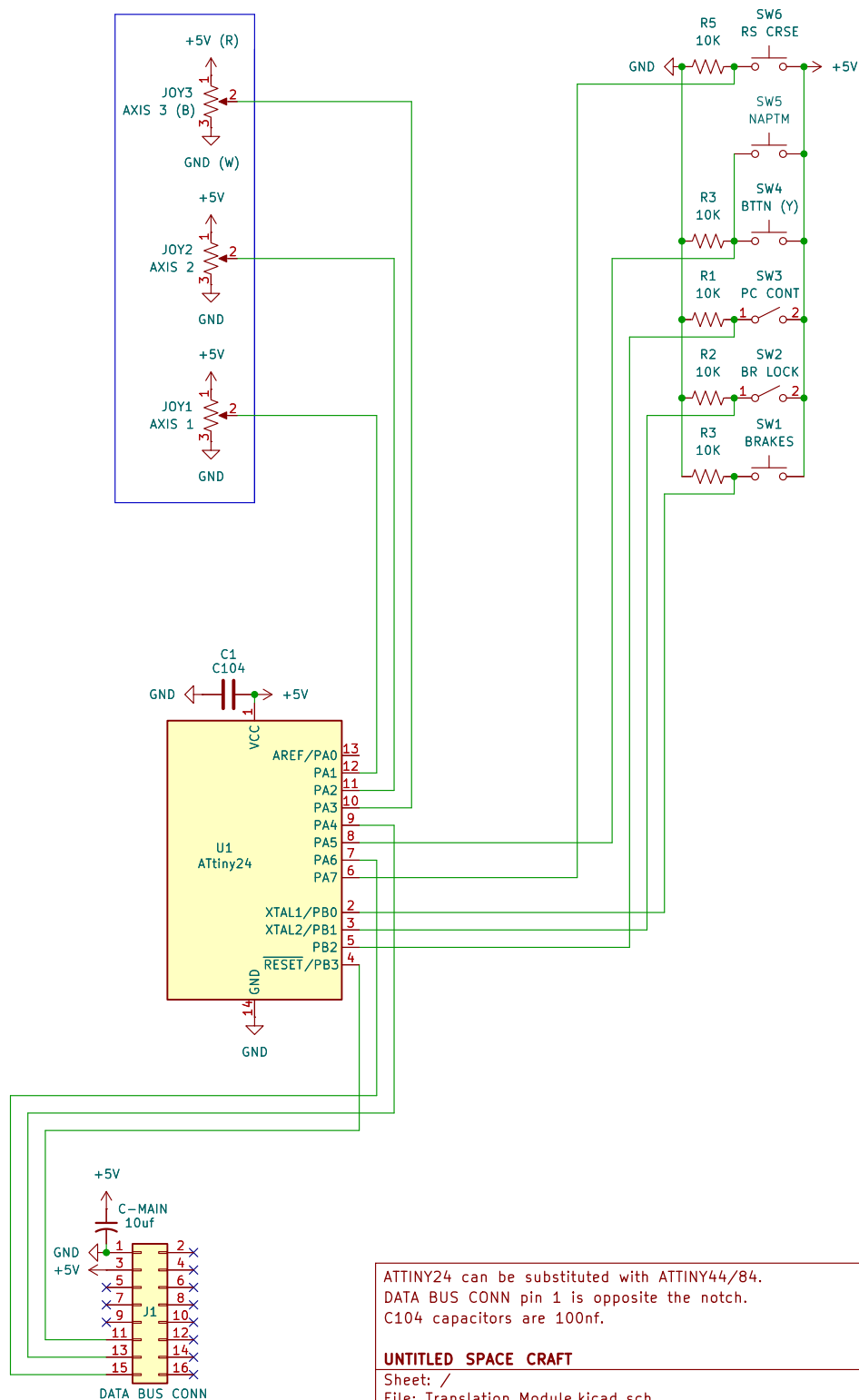
Title: TELEMETRY MODULE

Size: A4
Date: 2026-06-17
KiCad E.D.A. 10.0.3

Rev: 1.7
Id: 1/1







ATTINY24 can be substituted with ATTINY44/84.
DATA BUS CONN pin 1 is opposite the notch.
C104 capacitors are 100nf.

UNTITLED SPACE CRAFT

Sheet: /

File: Translation_Module.kicad_sch

Title: TRANSLATION MODULE

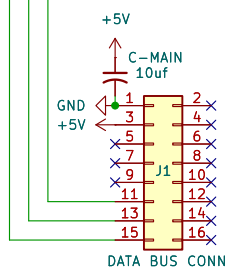
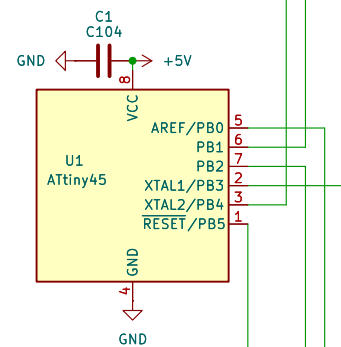
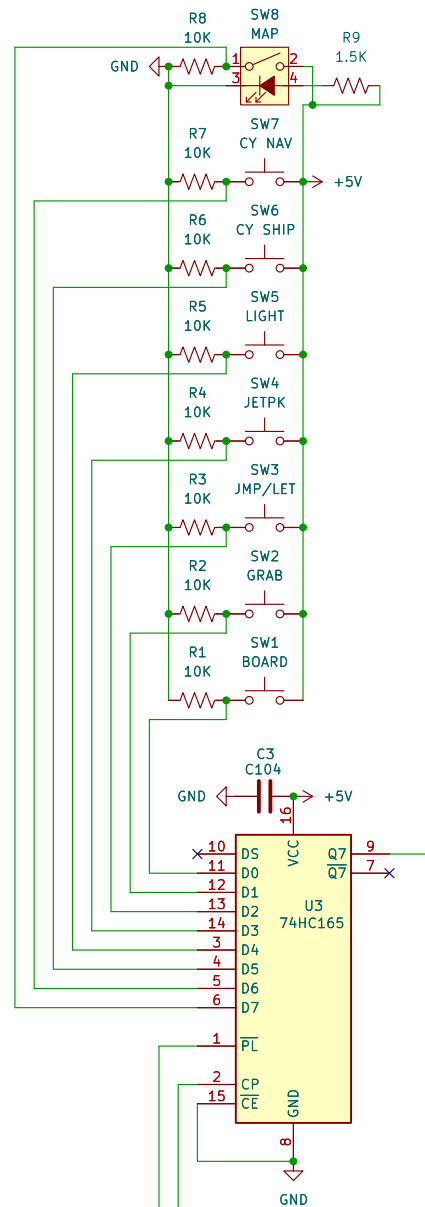
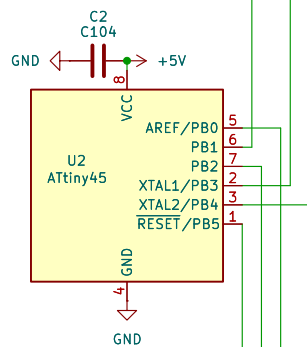
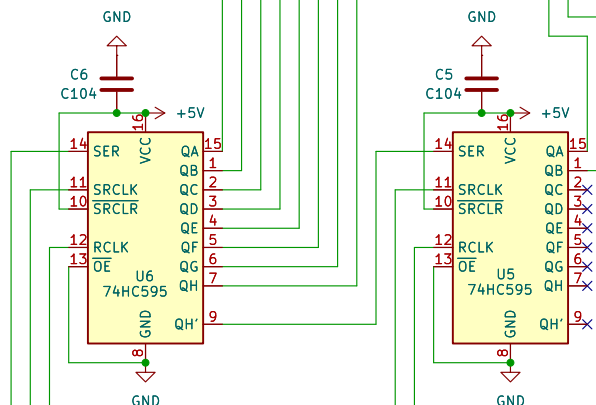
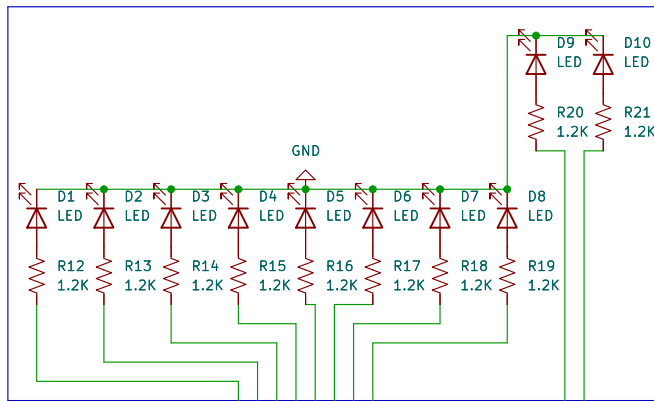
Size: A4

Date: 2026-06-17

Rev: 1.7

KiCad E.D.A. 10.0.3

Id: 1/1



ATTINY45 can be substituted with ATTINY85.
DATA BUS CONN pin 1 is opposite the notch.
C104 capacitors are 100nf.

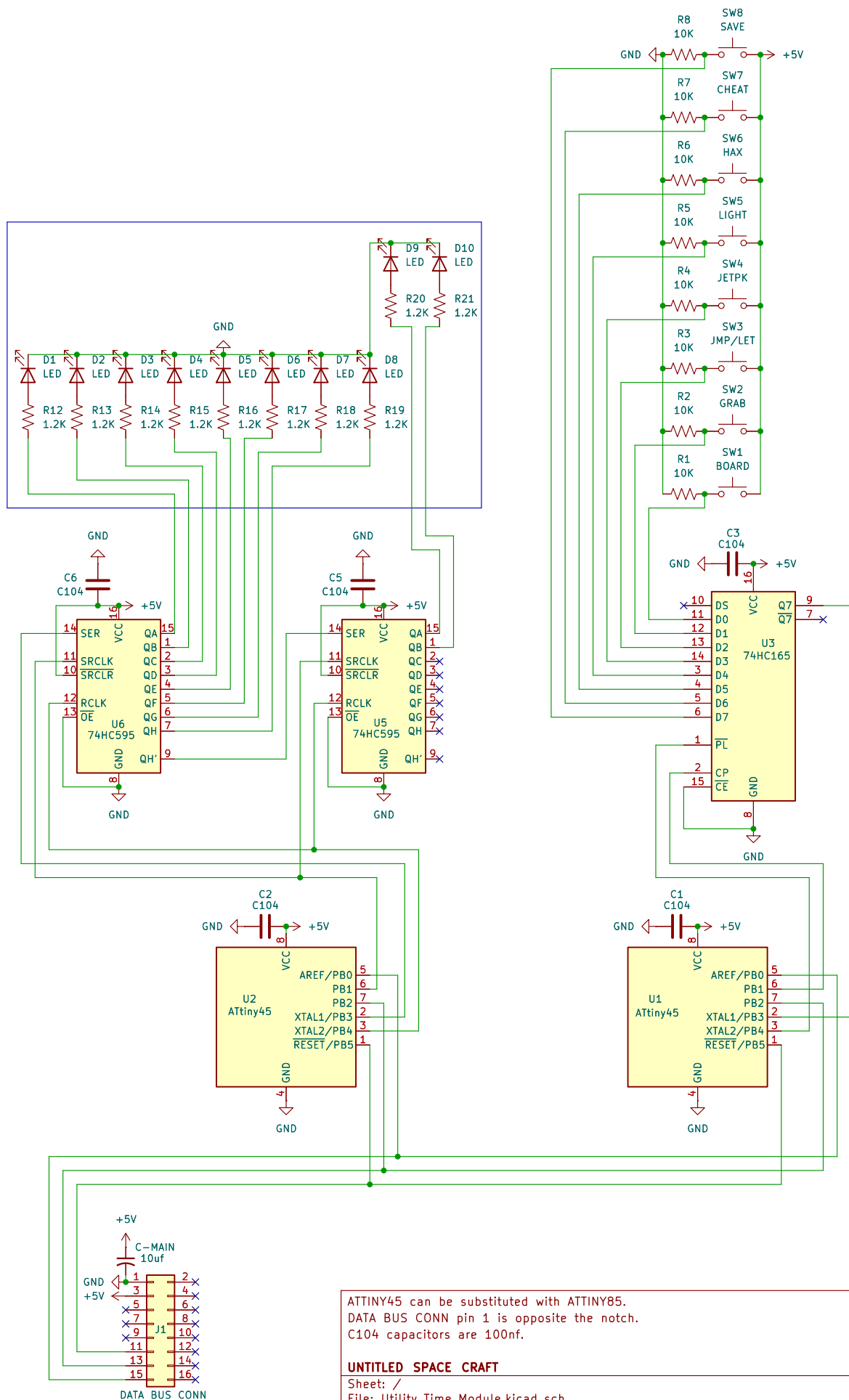
UNTITLED SPACE CRAFT

Sheet: /
File: Utility_Navigation_Module.kicad_sch

Title: UTILITY NAVIGATION MODULE

Size: A4 Date: 2026-06-17
KiCad E.D.A. 10.0.3

Rev: 1.7
Id: 1/1



ATTINY45 can be substituted with ATTINY85.
DATA BUS CONN pin 1 is opposite the notch.
C104 capacitors are 100nf.

UNTITLED SPACE CRAFT

Sheet: /
File: Utility_Time_Module.kicad_sch

Title: UTILITY TIME MODULE

Size: A4 Date: 2026-06-17
KiCad E.D.A. 10.0.3

Rev: 1.7
Id: 1/1